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Modeling and Control of Cable-Driven UAV Systems for Collaborative Load Manipulation

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Abstract

Acknowledgements

- My professors;
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Notations

$F_0(O, x_0, y_0, z_0)$	Inertial reference world frame
$F_p(P, x_p, y_p, z_p)$	Platform reference frame at its center of mass
$F_i(A_i, x_i, y_i, z_i)$	Quadrotor reference frame at cable connection
$F_{C_i}(A_{C_i}, x_{C_i}, y_{C_i}, z_{C_i})$	Quadrotor reference frame attached to the quadrotor at its center of mass
$F_k(A_k, x_k, y_k, z_k)$	Ground vehicle reference frame

Abbreviations

ACT	Aerial Cable Towed
ACTR	Aerial Cable Towed Robot
ACTS	Aerial Cable Towed System
AW	Available Wrench set
CDPR	Cable-Driven Parallel Robot
DKM	Direct Kinematic Model
DoF	Degree of Freedom
IKC	Inverse Kinematic Controller
INDI	Incremental Nonlinear Dynamic Inversion
MPC	Model Predictive Control
NMPC	Nonlinear Model Predictive Control
OCP	Optimal Control Problem
ORCA	Optimal Reciprocal Collision Avoidance
rAW	Radius of the Available Wrench
SAW	Static Available Wrench set
SW	Static workspace
UAV	Unmanned Aerial Vehicle
UGV	Unmanned Ground Vehicle
VIC	Variable Impedance Control
WFW	Wrench Feasible Workspace

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Chapter 1

Introduction

This thesis work takes place in the context of the ShieldBot project, which aims to revolutionize the construction, renovation, and maintenance sectors by investigating and validating a new generation of robotic systems tailored to the industry's unique challenges [1]. The ShieldBot project focuses on three complementary robotic solutions:

- InnerShieldBot is designed for internal construction works to enhance the thermal performance of walls and ceilings.
- InspectionShieldBot is dedicated to identify structural or insulation needs, and ensure long-term building integrity.
- FaçadeShieldBot is designed to operate on building façades and install thermal shielding in order to improve insulation and reduce energy consumption.

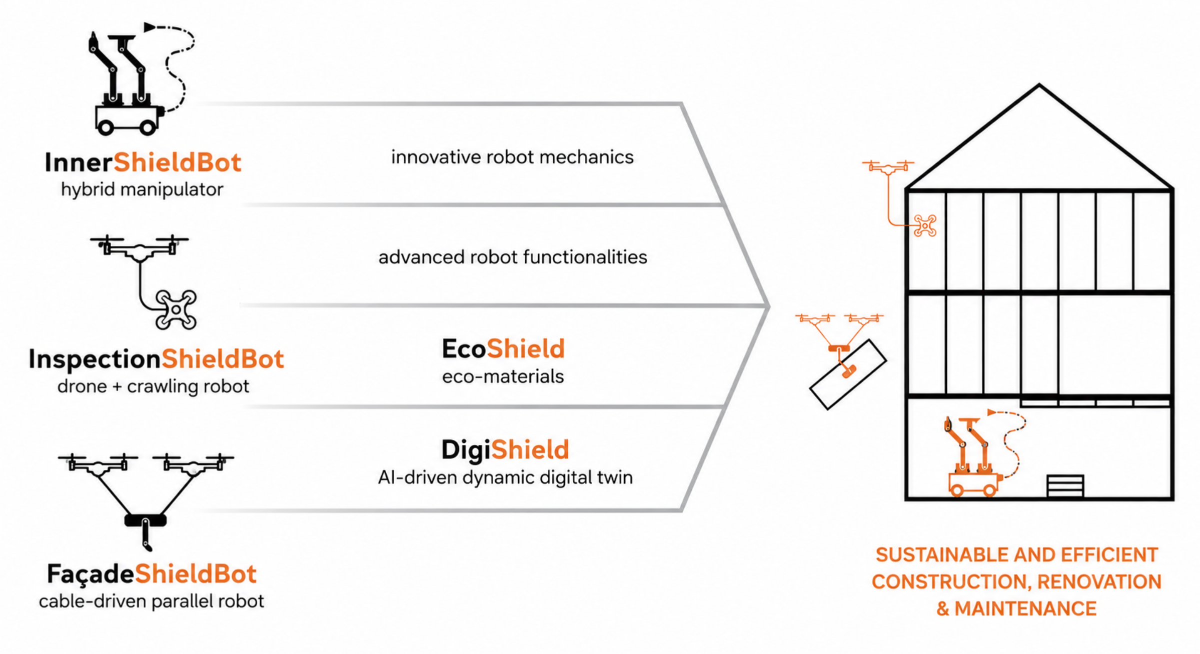


Figure 1.1: Robotic solutions developed by the ShieldBot project [1]

This thesis focuses on the robotic manipulation challenges inherent to the FaçadeShieldBot scenario. The system must generate sufficient force to manipulate the payload while maintaining a stable pose, as the application requires transporting and accurately positioning a payload near a vertical surface. At the same time, it must operate across large façades without requiring complex or permanent infrastructure on the building.

The overall system employs a platform-mounted robotic arm to manipulate and position tools along the surface. However, the scope of this work is strictly centered on the aerial platform itself and its ability to maintain a controlled pose without relying on permanent building infrastructure.

Previous robotic developments have demonstrated the potential of cable-based systems for construction applications. For example, the TitanBot project investigates cable-driven parallel robotic (CDPR) systems capable of transporting loads over large areas and reducing the physical effort required from construction workers [2]. Such systems benefit from the mechanical simplicity and large workspace offered by cable-based actuation. Additionally, their actuators can be located away from the manipulated payload, which makes them particularly attractive for applications involving heavy loads. However, conventional configurations generally rely on fixed or ground-based cable attachment points, which can limit their flexibility when operating over complex or changing environments.

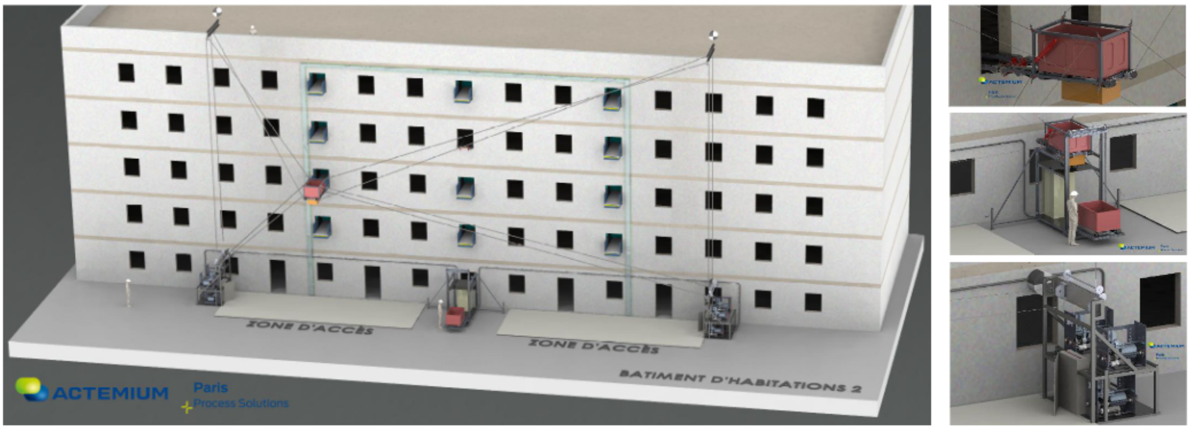


Figure 1.2: Solution proposed by TitanBot project [2]

Aerial robotic systems are a promising approach to access tight and difficult-to-reach areas while reducing the need for traditional scaffolding. Additionally, the use of cables introduces distance between the UAVs and the building, removing direct contact between the two (Figure 1.3). This is advantageous from a safety and operational perspective, as it reduces the risk of unintended collisions and the impact of the airflow generated by the rotors, allowing the system to safely come into contact with vertical surface [9].

Nonetheless, the position and orientation of the UAVs directly influence the forces and moments that can be generated through the cables. As a result, accurate positioning of the payload can become difficult, particularly when the system is required to perform precise manipulation close to a façade. These limits are especially critical in applications where the payload needs to be precisely positioned and then manipulated against a vertical surface.

The combination of aerial and ground-based actuation can improve the ability of the system to generate and control forces during interaction with the façade while retaining the flexibility of an aerial cable-towed system.

The resulting system, however, is characterized by a large set of design parameters that need to be selected at the design stage and remain fixed during the operation.

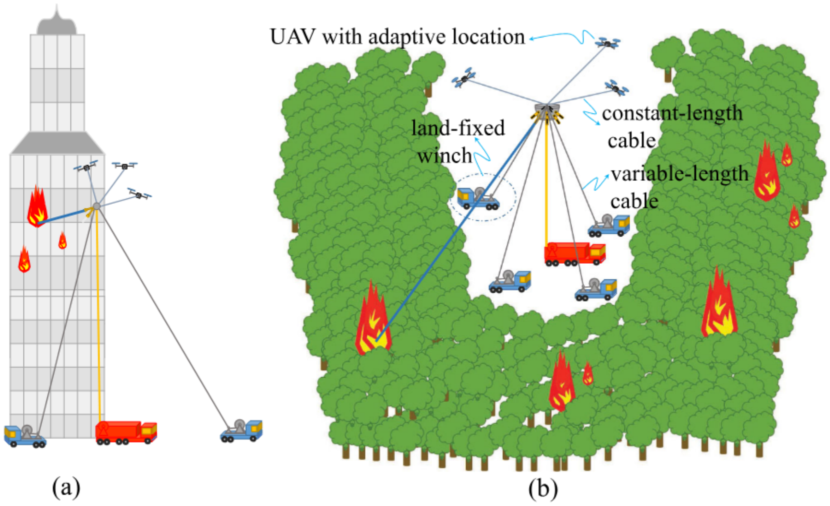


Figure 1.3: Using ACTRs with land-fixed winches for firefighting applications on (a) high-rise buildings and (b) vast areas such as forests [3]

They strongly influence the workspace, wrench feasibility, controllability, robustness, and energy efficiency of the system. The main design degrees of freedom include:

- The number of aerial and ground vehicles (UAVs and UGVs), which determines the level of actuation redundancy and force distribution capability;
- The number of cables and their routing, which affect the achievable wrench set and the system's stability;
- The cable length actuation strategy, such as fixed-length or winch-actuated cables;
- The system architecture and graph rigidity, including centralized and decentralized control configurations.

As a result, architectural design cannot be isolated from control system development. A configuration that is mechanically feasible may nevertheless provide poor force distribution or positioning performance in the region of interest. Conversely, an appropriate choice of architecture and attachment locations can improve the controllability and robustness of the system without necessarily increasing its mechanical complexity.

The main objective of this thesis is to study the modelling, design, and control of a hybrid Aerial Cable-Towed System (ACTS) for payload manipulation. In particular, the thesis focuses on how the architecture and cable configuration influence the ability of the system to manipulate a payload and generate the required payload wrench. The work focuses on the selection of suitable cable routing and attachment configurations and on the development of a control approach based on the resulting system model. The proposed configurations are evaluated through simulation to assess their performance and identify the most suitable architecture for the considered application.

The thesis is organized as follows.

Chapter 2

Chapter 3

Chapter 4

Chapter 5

Chapter 6
Chapter 7

Chapter 2

State of the Art

2.1 Cooperative Cable-Driven Aerial Systems

Several robotic architectures can be considered for the manipulation of a payload using multiple aerial vehicles. Among them, Cable-Driven Parallel Robots (CDPRs), Aerial Cable-Towed Systems (ACTSs), and Flying Parallel Robots (FPRs) are particularly relevant to the problem considered in this thesis. These architectures differ mainly in how the payload is connected to the actuators and consequently in their workspace, force-generation capabilities, and mechanical complexity.

2.1.1 Aerial Cable-Towed Systems

An Aerial Cable Towed Robot (ACTR) is a robotic system where the load is connected to multi-rotor UAVs through cables [4]. Other cables may be actuated also by non-aerial devices, such as actuators fixed to the ground or UGVs. In these systems, the payload motion is achieved by coordinating the cable tensions generated by the different actuators.

While standard ACTRs are limited in the directions and magnitudes of forces they can generate, the combination of mobile UAVs and fixed ground anchors enables the system to adapt its available wrench set to task requirements. This allows the platform to exert higher and more controllable interaction forces compared to configurations without a ground-based tension reference [3].

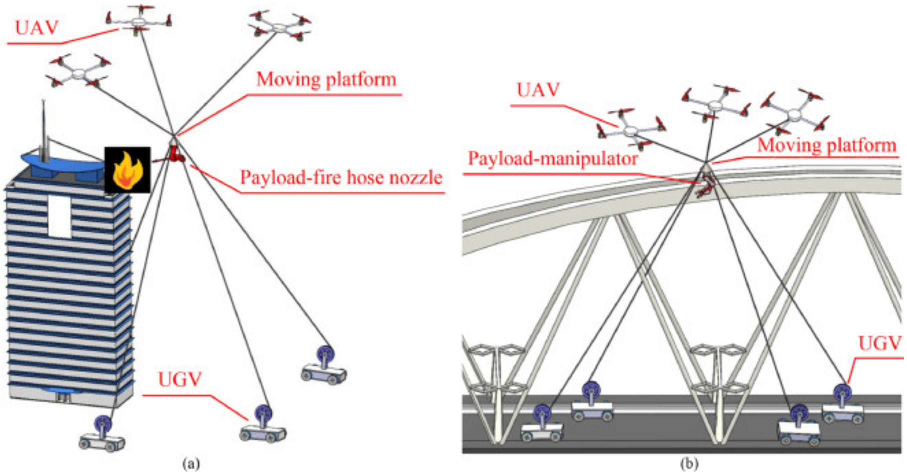


Figure 2.1: ACTRs with land-fixed winches for firefighting applications (a) and repair of a large bridge (b) [4]

2.1.2 Cable-Driven Parallel Robots

In conventional CDPRs, the end-effector is manipulated through flexible cables connected to fixed anchor points on the ground, and its motion is controlled by adjusting the cable lengths. Although CDPRs do not employ aerial robots, their analysis can be beneficial for making progress on ACTSs. As proposed by [9], ACTSs can be interpreted as reconfigurable CDPRs (RCDPRs), where fixed anchor points are replaced by mobile aerial platforms. For this reason, existing results on CDPR kinematics, dynamics, and control should not be ruled out a priori. A fundamental difference between CDPRs and ACTSs lies in the available tension space. In classical CDPR the tension's limits are fixed and do not depend on the robot's pose: the minimum tension is usually a small positive value to avoid slack, while the maximum tension is set by the motor torque. On the other hand, UAVs in ACTSs must always generate enough thrust to keep itself in the air and maintain its attitude, reducing the tension margin available for payload manipulation [10]. Therefore, CDPR findings can still be used for ACTSs, but they need to be adapted by introducing additional constraints due to the quadrotor actuation limits [9, 11].

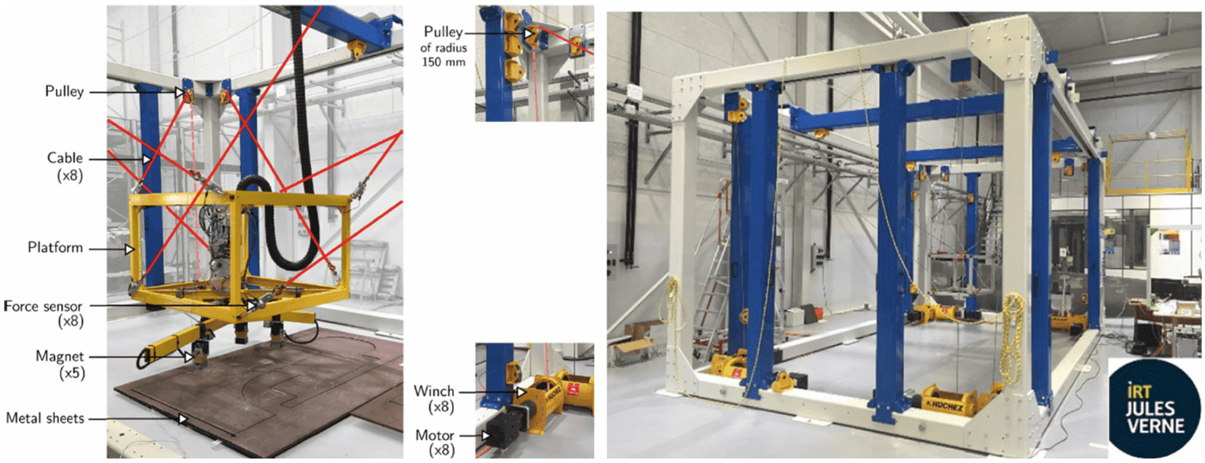


Figure 2.2: CDPR located at IRT Jules Verne [5, 6]

2.1.3 Flying Parallel Robots

Flying Parallel Robots (FPRs) changes from the previous two designs in that they connect the UAVs to the manipulated platform by rigid connections rather than cables, allowing both pushing and pull forces to be transmitted. The rigid legs also incorporate passive joints, providing additional passive degrees of freedom within the kinematic chains. These joints can reorient without actuation, preventing the platform from becoming over-constrained while maintaining the ability to transmit compressive loads [12]. This makes FPRs particularly well suited for tasks involving sustained physical interaction with the environment. However, the rigid links significantly increase the system's weight and mechanical complexity compared with cable-based architectures [7].

ACTSs achieve a similar interaction objective relying on coordinated control of multiple cable tensions. When ground anchors or UGVs are incorporated, the system can generate the reaction forces required for façade interaction while preserving the lightweight and mechanically simple nature of cable-driven robots.

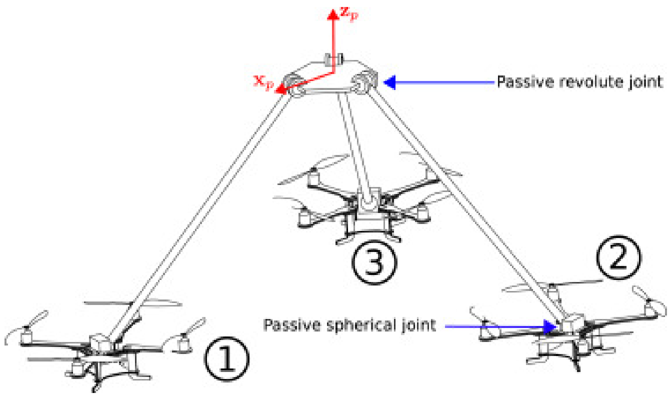


Figure 2.3: Kinematic scheme of the 3-DSR flying parallel robot [7]

2.2 Interaction with the Environment

Physical interaction is particularly challenging for aerial robotic systems because, unlike ground-based manipulators with a fixed base, aerial vehicles must simultaneously generate the forces required to remain airborne and those required to interact with the environment. Aerodynamic disturbances and modelling uncertainties further complicate the control problem [12]. In a cable-suspended system, these effects are coupled: a change in the position of one UAV or of the payload modifies the cable directions and consequently the forces experienced by the other agents [13].

These challenges become particularly important when the payload operates close to a vertical surface. Accuracy is a critical requirement in such applications, and interaction with the environment introduces additional force constraints that are not present during free-space motion [3]. Near-façade operation can also be affected by aerodynamic phenomena. Ground and ceiling effects modify the UAV dynamics, while the propeller wash generated by one UAV can disturb other vehicles operating in proximity [7, 13]. Furthermore, uncertainties in payload mass and inertia can affect the required thrust and lead to tracking errors or actuator saturation [9, 13].

Several forms of interaction have been considered for ACT systems, including unexpected collisions, planned contact, and active interaction with an external agent [14]. For the FaçadeShieldBot application, the relevant case is planned physical interaction with a vertical surface.

2.3 Control of Cable-Suspended UAV Systems

2.3.1 Control Architecture

Control architectures for cooperative aerial transportation can broadly be classified as centralized, distributed, or decentralized. In a centralized architecture, information about the UAVs, payload, and system configuration is available to a common controller. This allows the controller to coordinate the agents and explicitly account for the coupling between their motions.

A common approach is to model the ACTS as a continuously reconfigurable cable-driven parallel robot. In this framework, the controller can coordinate the payload motion, cable tensions, and UAV motion within a single control architecture [11, 9].

In [11], the design of a centralized trajectory tracking controller is proposed that explicitly accounts for both the kinematics and dynamics of the system. The control architecture is structured into three main layers:

- A task-space control loop that computes the desired wrench acting on the payload;
- A tension distribution algorithm that determines a feasible set of cable tensions realizing that desired wrench; and
- A joint-space control loop that converts the required cable forces into thrust and attitude commands for each quadrotor.

In contrast, distributed control architectures do not rely on a single controller. Instead, each UAV computes its control action using locally available information while exchanging information with neighbouring agents to achieve a common objective. Consensus algorithms are commonly employed to estimate global quantities, such as the payload state or formation geometry, without requiring complete knowledge of the system [15]. Decentralized control assumes that each UAV relies solely on its own local measurements, without explicit information exchange with the other agents. In this case, preserving the formation geometry becomes critical.

The system considered here is assumed to operate under centralized control, for which the states and configuration of the relevant agents are available to a common controller.

2.3.2 Cable Tension and Force Control

Cable tension management is fundamental to the control of cable-driven systems because cables can only transmit tensile forces. The controller must therefore ensure that the tensions remain positive to prevent cable slackness and maintain the desired mechanical configuration.

In redundant cable-driven systems, several sets of cable tensions can generate the same net wrench on the payload. This redundancy can be exploited to satisfy additional constraints while maintaining the desired payload wrench. In particular, internal forces corresponding to the null space of the wrench mapping can be used to modify the tension distribution without changing the resulting payload wrench [11, 3].

For ACTSs, tension feasibility is further constrained by the UAVs' actuation limits. Each UAV must allocate part of its available thrust to compensate for its own weight and maintain its attitude, leaving only a limited portion of its actuation capability for payload manipulation [10]. A valid control solution must therefore consider both the desired payload wrench and the available actuation capabilities.

More recent approaches incorporate cable tension constraints directly into trajectory optimization, allowing future trajectories to be evaluated according to their feasibility with respect to cable tension and actuator limits [13]. Such approaches demonstrate the importance of considering cable forces explicitly when planning or controlling cooperative aerial systems.

2.4 Positioning of this work in the current state of the art

Existing research on ACTS has focused primarily on modeling, trajectory generation, and controller design. In contrast, little attention has been paid to the systematic design of the architecture itself. In particular, selecting the number and placement of cable attachment points and choosing appropriate cable routes. These decisions directly influence the set of achievable torques, controllability, workspace, robustness, and the performance of any controller.

This thesis addresses these challenges by proposing a methodology for the design and evaluation of aerial cable-towing systems (ACTS) that combine UAVs and ground anchors. First, a general kinematostatic model is developed to describe ACTS architectures. Based on this model, a comprehensive preselection study is introduced to identify feasible cable configurations that satisfy the constraints of wrench closure, wrench feasibility, and interference constraints. The candidate architectures are then evaluated using both geometric performance indices and closed-loop dynamic simulations employing the proposed control framework.

Particular attention is given to the positioning of UAV attachment points and the arrangement of ground anchors, providing a quantitative comparison of various possible configurations. Thanks to the proposed methodology, it will be possible to design, evaluate, and select ACTS architectures, bridging the gap between the theoretical analysis of cable-controlled robots and the practical implementation of collaborative aerial manipulation systems.

Chapter 3

ACTS Design and Modeling

3.1 System Architecture

To formulate a general methodology, it is important to adopt a parameterization of the ACTS. We consider an ACTS with n_q quadrotors, n_g ground vehicles, m cables, and a load with d degrees of freedom. The i-th quadrotor is connected to the load at point B_i through a cable of length l_i, whose other end is fixed to the vehicle at point A_i, for $i = 1, \dots, n_q$. Similarly, the j-th ground vehicle is connected to the load at point B_j via a cable of length l_j, attached to the vehicle at point A_j, for $j = 0, \dots, n_g$. To avoid ambiguity between aerial and ground actuators, alphabetical indices may be used in the latter case to distinguish the two sets of vehicles [10].
We can also define the following systems of reference:

- $F_0(O, x_0, y_0, z_0)$ as the inertial reference world frame;
- $\mathbf{F}_p(P, x_p, y_p, z_p)$ as the frame attached to the platform at its center of mass;
- $F_i(A_i, x_i, y_i, z_i)$ as the frame attached to the quadrotor at point A_i, with the z-axis normal to the plane of the rotors;
- $F_{C_i}(A_{C_i}, x_{C_i}, y_{C_i}, z_{C_i})$ as the frame attached to the quadrotor at its center of mass A_{C_i} with the x-axis pointing forward, the y-axis pointing left and the z-axis pointing upwards;
- $F_k(A_k, x_k, y_k, z_k)$ as the frame attached to the ground vehicle at point A_k, with the z-axis normal to the ground.

In particular, since the ground vehicles are considered fixed in pose, also their reference frame is inertial and the fixed in time transformation matrix T_k from the reference world frame to each ground vehicle is known.

In this thesis, the following assumptions are adopted:

- The cables are modeled as massless, non-extensible and perfectly straight [14, 10, 9];
- The cable attachment point coincides with the quadrotor's center of mass [7, 10, 12];
- There are no aerodynamic disturbances, such as the rotor wash [7, 12, 15];
- The aerodynamic friction and resistance are neglected [16, 10];

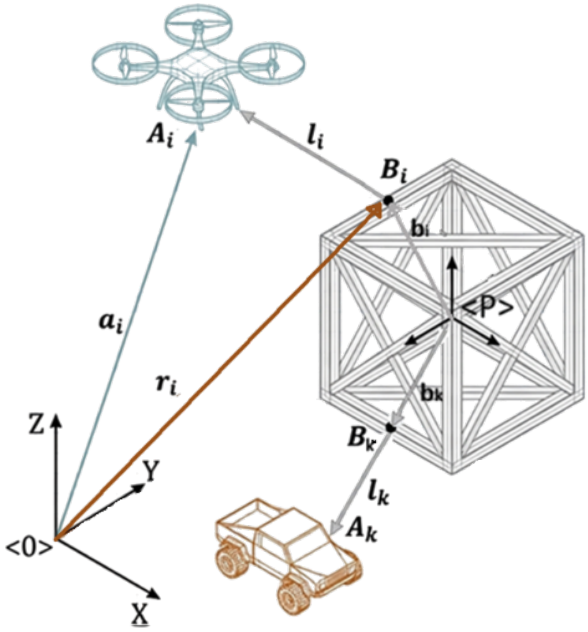


Figure 3.1: Graphical representation of the platform

The analyses presented in this chapter assume quasi-stationary operation of the ACTS. Under this assumption, the payload velocities and accelerations are sufficiently small that inertial effects can be neglected with respect to gravity and cable tensions. Consequently, the system is modelled in static equilibrium, allowing the architecture to be evaluated through its wrench-generation capabilities. The validity of this assumption is consistent with the intended façade interaction tasks, which involve slow and controlled payload motions. Dynamic effects are subsequently assessed through simulation in Chapter 6.

Lastly, the ground vehicles will be considered fixed anchoring points, as the ShieldBot project is still at its early stages.

3.2 Kinematostatic model

As talked about in Section 2.1.2, it is possible to model cable-suspended UAV systems as reconfigurable cable-driven parallel robots (RCDPRs), where the traditional fixed actuators are replaced by mobile UAVs, possibly integrated with land-fixed winches to balance interaction forces at high altitudes.

A kinematostatic model is a model that combines kinematics (motion and geometry) with statics (forces and equilibrium) to describe how a mechanical system moves and how forces are transmitted through it.

3.2.1 Kinematic model

What is the convention for bold vs normal characters?

Given the payload twist ${}^0\boldsymbol{\xi} = \begin{bmatrix} {}^0\mathbf{v}_p \\ {}^0\boldsymbol{\omega}_p \end{bmatrix} \in \mathbb{R}^6$ expressed in the inertial reference frame $< 0 >$, we can write the position and velocity of the attachment point B_i (resp. B_k) of the aerial vehicle (resp. ground) in $< 0 >$.

$$ {}^{0}\mathbf{r}_i = {}^{0}\mathbf{p} + {}^{0}\mathbf{R}_P\, {}^{P}\mathbf{b}_i, \tag{3.1} $$

where ${}^{P}\mathbf{b}_i$ is the position of the attachment point relative to the payload frame $< P >$. The corresponding velocity of the attachment point expressed in the inertial frame is

$$ {}^{0}\dot{\mathbf{r}}_i = {}^{0}\mathbf{v}_p + {}^{0}\boldsymbol{\omega}_p \times \left({}^{0}\mathbf{R}_P\, {}^{P}\mathbf{b}_i\right). \tag{3.2} $$

For definition, the cable vector is $\mathbf{l}_i = \mathbf{a}_i - \mathbf{r}_i$ where $\mathbf{a}_i$ is the anchor position and $\mathbf{r}_i$ is the corresponding payload attachment-point position. Hence, $\mathbf{l}_i$ points from B_i towards A_i. Under a centralized controller, the positions of all UAV A_i and UGV A_k are known which directly determine the anchor vector $\mathbf{a}_i$. Furthermore, the payload's position and orientation are assumed to be known together with the known locations of the attachment points relative to the payload, this information allows us to calculate the cable vectors $\boldsymbol{l}_i$.

The cable length and unit cable direction are given by

$$ l_i = \|\mathbf{l}_i\|, \qquad \mathbf{u}_i = \frac{\mathbf{l}_i}{\|\mathbf{l}_i\|} = \frac{\mathbf{a}_i - \mathbf{r}_i}{\|\mathbf{a}_i - \mathbf{r}_i\|} \tag{3.3} $$

The final form of cable vector variation is given by applying Equation 3.2 in its definition:

$$ \dot{\mathbf{l}}_i = \dot{\mathbf{a}}_i - \dot{\mathbf{r}}_i = \dot{\mathbf{a}}_i - \left(\mathbf{v}_p + \boldsymbol{\omega}_p \times \left({}^{0}\mathbf{R}_P\, {}^{P}\mathbf{b}_i\right)\right). \tag{3.4} $$

Since $l_i = \|\mathbf{l}_i\|$, the cable-length rate is $\dot{l}_i = \mathbf{u}_i^\top \dot{\mathbf{l}}_i$. Each $\dot{l}_i$ corresponds to the projection of the relative velocity of B_i along the cable direction

$$ \begin{aligned} \dot{l}_i &= \mathbf{u}_i^\top \left(\dot{\mathbf{a}}_i - \left(\mathbf{v}_p + \boldsymbol{\omega}_p \times \left({}^{0}\mathbf{R}_P\, {}^{P}\mathbf{b}_i\right)\right)\right) \\ &= \mathbf{u}_i^\top \dot{\mathbf{a}}_i - \begin{bmatrix} \mathbf{u}_i^\top & \left({}^{0}\mathbf{R}_P\, {}^{P}\mathbf{b}_i \times \mathbf{u}_i\right)^\top \end{bmatrix} \boldsymbol{\xi} \\ &= \mathbf{J}_{a,i}\dot{\mathbf{a}}_i - \mathbf{J}_{p,i}\boldsymbol{\xi}, \end{aligned} \tag{3.5} $$

where $\mathbf{J}_{a,i} = \mathbf{u}_i^\top \in \mathbb{R}^{1\times 3}$, and $\mathbf{J}_{p,i} = \begin{bmatrix} \mathbf{u}_i^\top & \left({}^{0}\mathbf{R}_P\, {}^{P}\mathbf{b}_i \times \mathbf{u}_i\right)^\top \end{bmatrix} \in \mathbb{R}^{1\times 6}$.

Let $m = n_a + n_g$ be the total number of winches (or cables) and $\mathbf{u}_i = l_i/\|l_i\|$ the unit cable direction. Define the cable-length vector as $\boldsymbol{\rho} = \begin{bmatrix} l_1 & l_2 & \cdots & l_m \end{bmatrix}^\top$ and the anchor position vector as $\mathbf{a} = \begin{bmatrix} \mathbf{a}_1^\top & \mathbf{a}_2^\top & \cdots & \mathbf{a}_m^\top \end{bmatrix}^\top \in \mathbb{R}^{3m}$.

The anchor Jacobian is

$$ \mathbf{J}_a = \begin{bmatrix} \mathbf{u}_1^\top & \mathbf{0}^\top & \cdots & \mathbf{0}^\top \\ \mathbf{0}^\top & \mathbf{u}_2^\top & \cdots & \mathbf{0}^\top \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0}^\top & \mathbf{0}^\top & \cdots & \mathbf{u}_m^\top \end{bmatrix} \in \mathbb{R}^{m\times 3m}. \tag{3.6} $$

he payload Jacobian is

$$ \mathbf{J}_p = \begin{bmatrix} \mathbf{J}_{p,1} \\ \vdots \\ \mathbf{J}_{p,m} \end{bmatrix} \in \mathbb{R}^{m\times 6}, \tag{3.7} $$

or explicitly,

$$\mathbf{J}_p = \begin{bmatrix} \mathbf{u}_1^\top & \left({}^0\mathbf{R}_P\,{}^P\mathbf{b}_1 \times \mathbf{u}_1\right)^\top \\ \vdots & \vdots \\ \mathbf{u}_m^\top & \left({}^0\mathbf{R}_P\,{}^P\mathbf{b}_m \times \mathbf{u}_m\right)^\top \end{bmatrix}. \tag{3.8}$$

The resulting kinematic model is

$$\dot{\boldsymbol{\rho}} = \mathbf{J}_a\dot{\mathbf{a}} - \mathbf{J}_p\boldsymbol{\xi} = \mathbf{J}_a \begin{bmatrix} \dot{\mathbf{a}}^{(a)} \\ \dot{\mathbf{a}}^{(g)} \end{bmatrix} - \mathbf{J}_p\boldsymbol{\xi}. \tag{3.9}$$

Let the cable tension vector be $\boldsymbol{\tau} = \begin{bmatrix} \boldsymbol{\tau}_{\text{uav}} \\ \boldsymbol{\tau}_{\text{ugv}} \end{bmatrix} \in \mathbb{R}^m$, where each τ_i is a scalar cable tension. The control allocation matrix is $\mathbf{J}_p^\top$, which maps the cable tensions to the payload wrench: $\mathbf{W}_p = \mathbf{J}_p^\top\boldsymbol{\tau}$, where

$$\mathbf{W}_p = \begin{bmatrix} \mathbf{F}_p \\ \mathbf{M}_p \end{bmatrix} \in \mathbb{R}^6. \tag{3.10}$$

Given a desired payload wrench $\mathbf{W}_p^*$, the control allocation problem is

$$\mathbf{J}_p^\top\boldsymbol{\tau} = \mathbf{W}_p^*, \qquad \boldsymbol{\tau} > \mathbf{0}. \tag{3.11}$$

The contribution of the i-th cable to the payload wrench is

$$\mathbf{W}_{p,i} = \mathbf{J}_{p,i}^\top\tau_i = \begin{bmatrix} \tau_i\mathbf{u}_i \\ \left({}^0\mathbf{R}_P\,{}^P\mathbf{b}_i\right) \times (\tau_i\mathbf{u}_i) \end{bmatrix}. \tag{3.12}$$

Consequently,

$$\mathbf{w}_p = \sum_{i=1}^{m} \mathbf{w}_{p,i} = \mathbf{J}_p^\top\boldsymbol{\tau}, \tag{3.13}$$

and the total force acting on the payload is

$$\mathbf{F}_p = \sum_{i=1}^{m} \tau_i\mathbf{u}_i. \tag{3.14}$$

3.2.2 Static model

Given the cable tension vector $\boldsymbol{\tau}$, the total cable power can be written as:

$$\begin{aligned} \underbrace{\boldsymbol{\tau}^\top\dot{\boldsymbol{\rho}}}_{\text{total cable power}} &= \underbrace{\boldsymbol{\tau}^\top\mathbf{J}_a\dot{\mathbf{a}}}_{\text{UAVs and UGVs mechanical power}} - \underbrace{\boldsymbol{\tau}^\top\mathbf{J}_p\boldsymbol{\xi}}_{\text{payload power}} \\ &= \left(\mathbf{J}_a^\top\boldsymbol{\tau}\right)^\top\dot{\mathbf{a}} - \left(\mathbf{J}_p^\top\boldsymbol{\tau}\right)^\top\boldsymbol{\xi} \\ &= \mathbf{F}_a^\top\,\dot{a} - \mathbf{W}_p^\top\,\boldsymbol{\xi} \end{aligned} \tag{3.15}$$

Each cable i applies a force on the payload $\mathbf{F}_{a,i}$ and an equal and opposite force on the UAV (resp. UGV). We can write balance equation on our elements:

Payload Balance

$$\underbrace{\boldsymbol{W}_p}_{\mathbf{J}_p^\top \tau} + \underbrace{\boldsymbol{W}_g}_{\begin{bmatrix} 0 \\ 0 \\ -m_p g \\ 0 \\ 0 \\ 0 \end{bmatrix}} + \boldsymbol{W}_{contact} + \underbrace{\boldsymbol{W}_{wind}}_{\text{let } = 0} = \mathbf{0} \tag{3.16}$$

In particular, $\boldsymbol{W}_{contact} = 0$ when there is no contact with the environment.

UAV balance

$$-\boldsymbol{F}_{a,i} + \boldsymbol{F}_{prop,i} + \underbrace{\boldsymbol{F}_{g,i}}_{\begin{bmatrix} 0 \\ 0 \\ -m_i g \end{bmatrix}} = \mathbf{0} \tag{3.17}$$

with ${}^i\boldsymbol{F}_{prop} = \begin{bmatrix} 0 & 0 & F_{prop,z,i} \end{bmatrix}^\top$ and $\mathbf{F}_{a,i} = \mathbf{J}_{a,i}^\top \tau_i = \tau_i \mathbf{u}_i$.

UAVs must always generate sufficient thrust to maintain flight and ensure their own stability. As a result, only a portion of their available thrust can be allocated to payload manipulation, which reduces the effective force that can be transmitted through the cables.

UGV balance

$$-\boldsymbol{F}_{a,j} + \underbrace{\boldsymbol{F}_{ground,j}}_{\begin{bmatrix} F_{frict,x,j} \\ F_{frict,y,j} \\ \mathbf{F}_{norm} \end{bmatrix}} + \underbrace{\boldsymbol{F}_{g,j}}_{\begin{bmatrix} 0 \\ 0 \\ -m_j g \end{bmatrix}} = \mathbf{0} \tag{3.18}$$

with $\|\mathbf{F}_{frict}\| \leq \mu \mathbf{F}_{norm}$, $\mathbf{F}_{norm} = m_j \mathbf{g} - \mathbf{F}_{a,z,j}$ and $F_{a,j}$ is obtained through Equation 3.19.

Similarly, UGVs are subject to ground interaction constraints. In particular, they must satisfy friction limits to remain stationary when no motion is commanded. This imposes bounds on the admissible cable forces, as excessive tension may cause slipping. By Newton's third law, the force the cable exerts on the drone $\mathbf{F}_{a,i}$ (resp. on the UGV $\mathbf{F}_{a,j}$) is equal and opposite to the force it exerts on the payload. Since $\mathbf{u}_i$ points from B_i toward A_i, the cable pulls the drone with

$$\mathbf{F}_{a,i} = \mathbf{J}_{a,i}^\top \tau_i = \tau_i \mathbf{u}_i \tag{3.19}$$

As a consequence, the total cable force on the payload is:

$$\mathbf{F}_p = -\sum_{i=1}^{n_a} \mathbf{F}_{a,i} - \sum_{j=n_a+1}^{n_a+n_g} \mathbf{F}_{a,j} \tag{3.20}$$

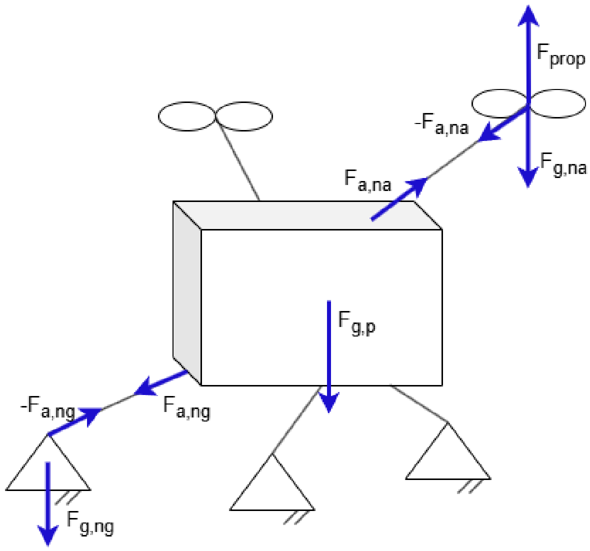


Figure 3.2: Forces acting on the ACTS

3.2.3 Kinematostatic model

Writing Equation 3.16, 3.17 and 3.18 in terms of τ, we obtain:

$$\mathbf{W}_{sys} = \begin{bmatrix} \mathbf{W}_p \\ \mathbf{F}_a \end{bmatrix} = \begin{bmatrix} \mathbf{J}_p^\top \\ \mathbf{J}_a^\top \end{bmatrix} \boldsymbol{\tau} = \mathbf{G}\boldsymbol{\tau} = \begin{bmatrix} -\mathbf{W}_g - \mathbf{W}_{contact} - \mathbf{W}_{wind} \\ \mathbf{F}_{prop} + \mathbf{F}_{ground} + \mathbf{F}_g \end{bmatrix} \tag{3.21}$$

where $\mathbf{F}_{prop,j} = 0$ for the UGVs and $\mathbf{F}_{ground,i} = 0$ for the UAVs. Then $\mathbf{W}_{sys} \in \mathbb{R}^{(6+3m)\times 1}$, $\mathbf{G} \in \mathbb{R}^{(6+3m)\times m}$, and

$$\mathbf{G}\boldsymbol{\tau} + \begin{bmatrix} \boldsymbol{W}_g + \boldsymbol{W}_{contact} + \boldsymbol{W}_{wind} \\ -\boldsymbol{F}_{prop} - \boldsymbol{F}_{ground} - \boldsymbol{F}_g \end{bmatrix} = 0 \tag{3.22}$$

3.3 Architecture Design

The configuration of a Aerial Cable Towed System (ACTS) has a direct influence on its workspace, force-generation capabilities, controllability, and energy consumption. For a given payload pose, multiple equilibrium configurations may exist, corresponding to different distributions of internal cable forces that do not modify the resulting net wrench acting on the payload. Although these configurations satisfy the same static equilibrium conditions, they can require significantly different UAV positions and cable tensions, leading to different energy consumption and robustness properties [17]. Consequently, the architecture of the ACTS must be selected according to the requirements of the intended application.

3.3.1 Design Variables

Each configuration is characterized by parameters such as the number and arrangement of aerial and ground actuator, the cable attachment points, and the cable routing. More specifically, the ACTS consists of at least one UAV, required as the payload needs to be lifted from the ground, and one UGV, included for safety and operational constraints.

Lastly, for a rigid payload with six degrees of freedom, at least six independent cable are required to generate a general six-dimensional payload wrench.
In this thesis, we suppose working with nine actuators, three aerial and six on the ground. Since we are interested in working with an over-actuated system, in this thesis we suppose working with nine actuators, three aerial and six on the ground. While the latter are used to fully control the payload pose, the three drones will act as an anti-gravitational force. A schematic is shown in Figure 3.3. The candidate architectures are subsequently evaluated using the performance indices introduced in Section 3.3.2.

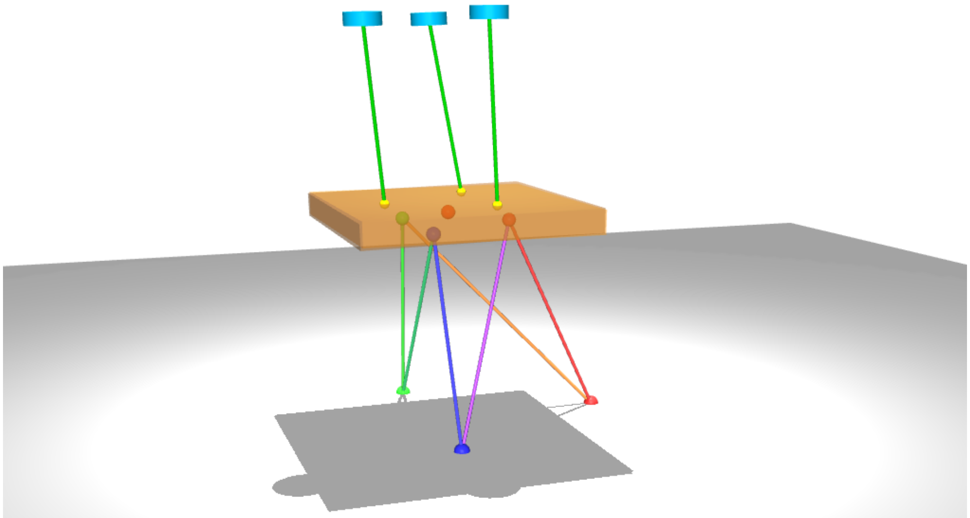


Figure 3.3: Over actuated ACTS with 6 ground cables and 3 UAVs

Aerial Actuators

The configuration of the UAV connections is crucial, as it directly affects how forces are applied to the payload: UAVs are responsible for generating the lifting force required to lift the payload. Consequently, their number depends strongly on the payload mass. However, an increased number of UAVs does not necessarily improve the performance of the system. Additionally, each UAV can be connected to the payload using either a single cable or multiple cables. The choice of configuration affects controllability, wrench feasibility, robustness, and energy efficiency and should be evaluated accordingly. A practical compromise is to use three UAVs connected to the payload at distinct attachment points, providing a balance between the metrics just mentioned (Figure 3.4).

As we will see in later discussions, the main alternatives considered in this work are illustrated in Figure 6.3.

Ground Actuators and Anchors

In contrast, UGVs do not lift the payload, so the same stability arguments are not valid. If we treat the UGVs as fixed anchor points, connecting a single UGV to multiple hook points on the payload simplifies to a standard fixed-anchor scenario. Because each actuator operates independently, this configuration adds no extra mathematical complexity. However, if the UGVs are actively moving, multi-cable connections require precise winch coordination to ensure all cables remain taut at all times. In Figure 3.5, possible payload hook attachments can be found.

The number and spatial distribution of these anchors affect the directions of the corresponding cable forces and therefore the wrench that can be generated at the payload. Different anchor layouts can consequently lead to different force and moment capabilities.

Payload Attachment Points

The locations of the cable attachment points on the payload directly affect the moments generated by cable tensions. Attachment points that are more widely separated can provide larger moment arms, whereas closely spaced attachment points result in different wrench and conditioning properties.

The attachment points are therefore treated as design variables in the architecture selection process.

Ground Anchor Placement and Cable Routing

The positions of the ground anchors determine the directions of the ground cables for a given payload pose. Their arrangement consequently affects both the achievable wrench and the distribution of cable tensions.

In addition to anchor placement, the routing of the cables between the ground anchors and the payload attachment points is considered as a design variable. Candidate routings include crossed and uncrossed arrangements, as well as adjacent and diagonal connections. These different routings can produce different cable directions and therefore different force-generation capabilities.

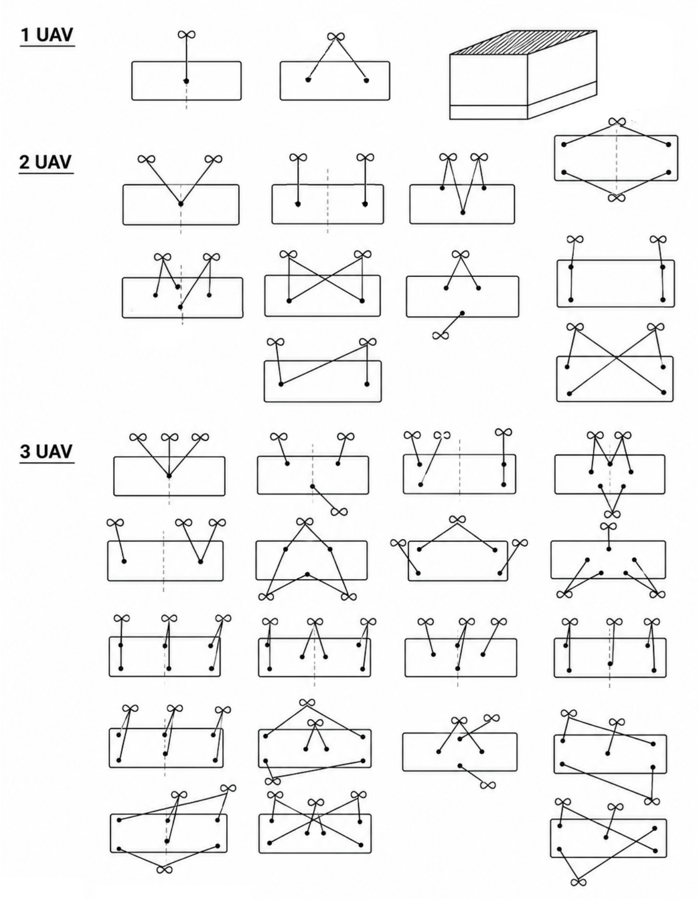


Figure 3.4: Possible UAVs connection to the payload

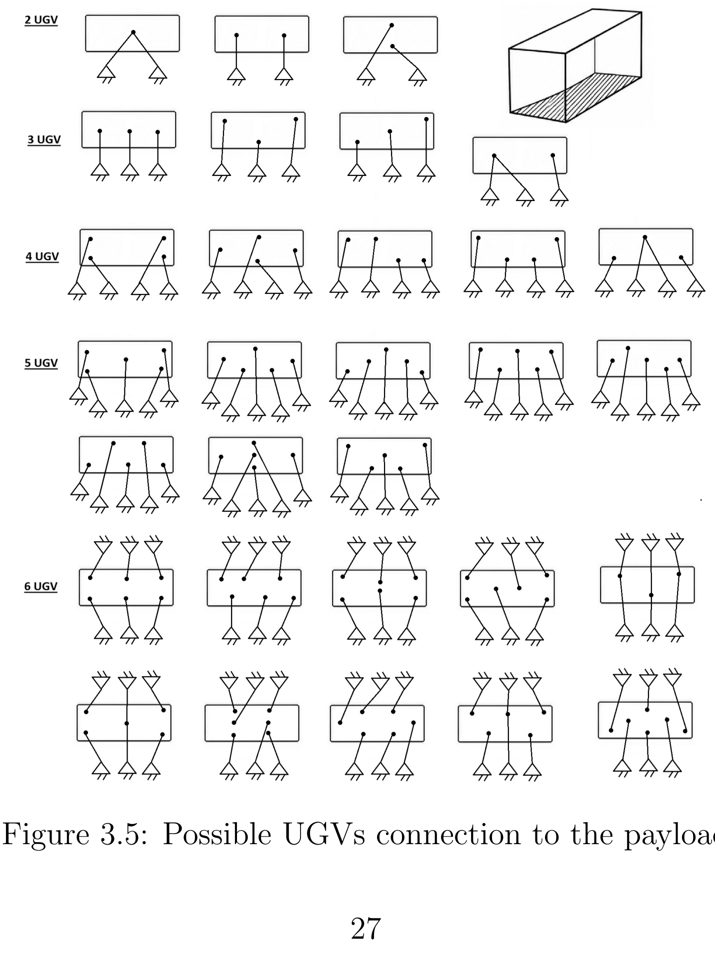


Figure 3.5: Possible UGVs connection to the payload

3.3.2 Performance Metrics

In the context of the ShieldBot project, we are working with two questions:

1. Which attachment geometry gives the best ability to generate the required payload wrench over the entire mission workspace?

2. Given the chosen architecture, where should the actuators (UAVs and anchor points in the ground) be positioned, and what cable lengths should be used during operation?

In both cases, we need an objective measurement that guides us in the selection.
Every configuration to be eligible needs to satisfy feasibility metrics first, such as [18]:

- Wrench closure conditions (WCC), which guarantees that the payload is fully constrained and arbitrary wrenches can be generated. Mathematically: $\text{rank}(\mathbf{J}_p) = 6$ and positive tensions exist.

- Wrench Feasible Condition (WFC), which checks whether tensions remain in the feasible bound $0 \leq \tau_{\min} \leq \tau \leq \tau_{\max}$, included UAV thrust limits, cable strength limits and UGV friction limits in a future where the ground anchor point will be movable.

- Interference-Free Condition (IFC), ensuring that there is no cable rubbing and no collision with payload edges. Mathematically:

$$\begin{aligned} d_{ij} &\geq d_{\text{safe}}, \quad \text{where } d_{ij} = \text{dist}\,(\text{cable } i, \text{cable } j) \qquad \forall i, j \\ d_i &\geq d_{safe}, \quad \text{where } d_i = \min_{s\in[0,1]} \left(\text{dist}\,(r_i + s(a_i - r_i), P)\right) \qquad \forall i \end{aligned} \tag{3.23}$$

  where P is the payload geometry and d_{safe} is a clearance threshold.

However, assuming that all cable attachment points are on the same face of the payload, the only feasible interference between the latter and one of the cables is a near-tangent cable that touches that face.

After passing the feasibility filters, we need a way to assign a score to each hook configuration.

Capacity margin or minimum degree of constraint satisfaction

The capacity margin is a robustness index that is used to analyze the degree of feasibility of a configuration. The feasibility region is the set of all poses and geometric configurations in which the system can physically balance a required task wrench, such as gravity or external interaction forces, while satisfying the quadrotor's actuation limits and the physical constraints of the cables [10]. It is defined as the shortest signed distance from the task wrench to the boundary of the available wrench set (AWS). Mathematically, it is the minimum of the distances of each one of the n vertices $w_{e,j}$ to each one of the p facets $(\mathbf{a}_l, b_l)$ of the AWS.

$$\gamma = \min_{j=1\div n} \min_{l=1\div p} \gamma_{j,l}, \quad \text{where } \gamma_{j,l} = \frac{b_l - w_{e,j}^\top \mathbf{a}_l}{\|\mathbf{a}_l\|_2} \tag{3.24}$$

where $\mathbf{a}_l$ is the normal vector to the l-th facet and b_l is its offset [19].

The wrench vector combines force components (N) and moment components (Nm), which are not directly comparable in a Euclidean sense. Computing the capacity margin γ by measuring facet distances in the raw, mixed-unit wrench space would make γ depend on the arbitrary choice of length unit used to express the cable attachment points, rather than on the physical robustness of the configuration. Following Ruiz et al. [19], the moment components of the wrench are therefore rescaled by a characteristic length.

$$r = \sqrt{\frac{1}{m}\sum_i \|b_i\|^2} \tag{3.25}$$

Where b_i is the position of the i-th cable attachment point relative to the payload center of mass, before the capacity margin is evaluated. This rescaling renders all six wrench components dimensionally homogeneous (in force units) and ensures γ is independent of the chosen length scale.

Worst-Case Capacity Margin

Additionally, we can define the worst-case capacity margin as to how far the system remains from actuator saturation in the most demanding operating condition. Mathematically:

$$\zeta = \min\left(1 - \max_i \frac{\tau_i}{\tau_{i,max}}\right) \tag{3.26}$$

Radius of Available Wrench (rAW)

The radius of the available wrench (rAW) is the maximum magnitude of force that the system can generate uniformly in all directions while remaining within admissible tension limits [3]. Its optimization helps identify arrangements that simultaneously maximize force robustness, geometric and collision-avoidance constraints, resulting into an enlargement of the static workspace [3].

$$rAW = \min_i\left(\min\left(\frac{|d_{pi}|}{\|\mathbf{n}_{null,i}\|}, \frac{|d_{qi}|}{\|\mathbf{n}_{null,i}\|}\right)\right) \tag{3.27}$$

being $u_{null,i} = \text{null}(M_i)$ with M_i containing combinations of linearly independent cable unit vectors, d_{pi} and d_{qi} defined as:

$$\begin{aligned} d_{pi} &= \sum_{j\in I^+} \tau_{max} u_{nulli}^T u_j + \sum_{j\in I^-} \tau_{min} u_{nulli}^T u_j - m|u_{nulli}^T g| \\ d_{qi} &= -\sum_{j\in I^-} \tau_{max} u_{nulli}^T u_j - \sum_{j\in I^+} \tau_{min} u_{nulli}^T u_j - m|u_{nulli}^T g| \end{aligned} \tag{3.28}$$

Since the formulation of rAW in [3] considers a point-mass payload, it does not account for the moments generated by the cable tensions about the payload's attachment points. To capture both effects, the radius of the available wrench is therefore split into two independent quantities: rAW$_{force}$, the maximum pure force the system can generate uniformly in any direction while producing zero net moment, and rAW$_{moment}$, the analogous quantity for pure moment generation with zero net force. Whereas rAW$_{force}$ admits the closed-form null-space expression of Eq. 3.27, rAW$_{moment}$ is instead obtained through a vertex-enumeration procedure: candidate cable tension vectors satisfying zero net force at the bounds of the admissible tension box are computed, their corresponding moments $\tau_i(\mathbf{b}_i \times \mathbf{u}_i)$ are collected, and rAW_{moment} is taken as the distance from the origin to the nearest facet of the convex hull of these moment vectors.

Conditioning index

The conditioning index measures how uniformly the system can generate forces and moments in different directions. Mathematically:

$$\nu = \frac{\sigma_{\min}(\mathbf{J}_p)}{\sigma_{\max}(\mathbf{J}_p)} \in [0, 1] \tag{3.29}$$

$\sigma_{\min}$ and $\sigma_{\max}$ are the minimum and maximum singular values of the Jacobian matrix, obtained via Singular Value Decomposition (SVD) [15]. A conditioning index close to one indicates a well-conditioned system, far-away from singular configuration.

Manipulability

The manipulability index, defined as

$$w = \sqrt{\det(\mathbf{J}_p \mathbf{J}_p^\top)} \tag{3.30}$$

measures the volume of the wrench ellipsoid. A value close to 0 means that the system hit a kinematic singularity, while large value means a large overall wrench generation capability.

Composite score

Once the feasibility constraints are satisfied, we may want to proceed either by

$$N = w_1\hat{\gamma} + w_2\,\mathrm{r\hat{A}W} + w_3\hat{\zeta} \tag{3.31}$$

where all the metrics are scaled to $[0, 1]$,

$$U^* = \arg\max_{U}\,(N(U)) \tag{3.32}$$

3.3.3 Selection Strategy

The search begins by exhaustively enumerating all physically valid cable routings for each payload–ground layout pair while respecting the maximum cable capacity of every attachment point. Each routing is evaluated over the representative set of payload poses introduced in Section 3.3.2.

The ground-cable subsystem is first screened using the Wrench Closure Condition (WCC) and the Interference-Free Condition (IFC). Routings violating either criterion at any pose are immediately discarded. The surviving routings are ranked according to their worst-case conditioning index across all tested poses, and the highest-ranked routing is selected for each architecture. For the highest-ranked routing, the Wrench Feasibility Condition (WFC) is then evaluated using the nominal UAV positions, in which each UAV is initially placed directly above its payload attachment point:

$$\min_{\boldsymbol{\tau}} \left\| J_{p,\text{full}}\boldsymbol{\tau} - \mathbf{W}^* \right\|^2, \qquad \text{subject to} \qquad \boldsymbol{\tau}^*_{\min} \le \boldsymbol{\tau} \le \boldsymbol{\tau}^*_{\max}, \tag{3.33}$$

where $\mathbf{W}^*$ denotes the desired payload wrench and the tension bounds account for the admissible cable tensions, including the maximum tension imposed by the UAV thrust capabilities.

A configuration is considered wrench-feasible when the minimum residual of Eq. 3.33 is below a prescribed tolerance. If this does not happen, the UAV positions are re-optimized. The routing is accepted only if this repositioned configuration satisfies the WFC criterion for every tested pose.

An architecture is considered feasible if its highest-ranked routing satisfies the WCC, IFC, and joint WFC over all tested poses. The surviving architectures are subsequently evaluated using the performance indices introduced in Section 3.3.2. The conditioning index is used only to rank candidate routings because it is a geometry-based metric that can be computed directly from the payload Jacobian without solving the tension optimization problem. The remaining performance indices are evaluated only after the architecture selection process.

This strategy separates the inexpensive combinatorial screening of ground-cable routings from the more computationally demanding analysis of the complete aerial-ground system. It therefore reduces the number of configurations requiring full nonlinear evaluation while retaining the relevant UAV-dependent constraints for the final architecture selection.

Algorithm 1 Architecture Selection Strategy

```
for all payload–ground layout pairs do
    for all possible cable routing do
        for all test poses do
            Build J_{p,ground}
            Check WCC                                   ▷ rank(J_{p,ground}) = 6
            Check IFC                                   ▷ Eq. 3.23
            if WCC or IFC fails then
                Reject routing
                continue
            end if
        end for
        Compute the worst-case conditioning index
    end for
    Select routing with largest worst-case conditioning index
    Build complete Jacobian                             ▷ Eq. 3.8
    Evaluate WFC using nominal UAV positions            ▷ 0 ≤ τ_min ≤ τ ≤ τ_max
    if WFC fails then
        Optimize UAV positions
        Re-evaluate WFC
    end if
    Accept architecture if WFC is satisfied
end for
```

Chapter 4

Control Architecture

In the course of this thesis, we divided the control of the ACTS using two separated subsystems. While the ground winches adjust the length of the lower cables to control the position of the payload relative to the ground anchors, the UAVs control the magnitude and direction of the thrust generated by their propellers to maintain the ground cables taut.

The coordinated action of these two subsystems allows the payload to achieve the desired pose while satisfying the mechanical constraints of the system. This chapter describes the adopted control architecture and its implementation within the proposed ACTS.

4.1 Force control with a point-mass payload

Before going into details on the full consideration, let's analyze some simpler configurations.

4.1.1 Case: one drone connected to the ground through a cable

The goal is to design the drone force F_{prop} such as that the ground tension is controlled. Since only a tension in the cable direction is imposable, an arbitrary force f^* at the ground anchor can be achieved only by moving the drone, changing the cable direction u. Called a the drone's position and b the attachment point to the ground:

$$u = \frac{a-b}{\|a-b\|} \tag{4.1}$$

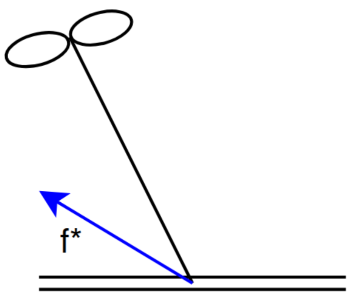


Figure 4.1: One drone system connected to the ground through a cable

Option 1

Defined the desired cable tension τ^* and the desired cable direction u^* as:

$$\tau^* = \|f^*\| \qquad (4.2) \qquad\qquad u^* = \frac{f^*}{\|f^*\|} \qquad (4.3)$$

The ground tension is obtained using the geometric constraint which relates the anchor point to the drone position:

$$a^* = b + l\, u^* \tag{4.4}$$

where l is the cable length. Equation 4.4 is then used as a reference to compute the propeller force.

$$F_{prop} = m_i g e_3 + K_p(a^* - a) + K_d(\dot{a}^* - \dot{a}) + \tau^* u \tag{4.5}$$

Option 2

Given Newton's Second Law applied to the drone described in Equation 3.17, we are able to estimate the cable force $\hat{f} = F_{a,i}^{est}$

$$m_i \ddot{a}_i = F_{prop,i} - \hat{f} + F_{g,i} \qquad \rightarrow \qquad \hat{f} = F_{prop,i} + F_{g,i} - m_i \ddot{a}_i \tag{4.6}$$

where all the forces are in the world frame and $\ddot{a}_i$ is the drone linear acceleration. Using Equation 3.17, we would like to apply a propulsion force equal to

$$F_{prop,i}^* = f^* - F_{g,i} \tag{4.7}$$

Finally, the closed loop equation is:

$$F_{prop,i} = F_{prop,i}^* + K_p e + K_d \dot{e}, \quad \text{with } e = f^* - \hat{f} \tag{4.8}$$

The result is given to the attitude controller, which aligns the body z-axis with the commanded thrust vector.

4.1.2 Case: payload connected to the ground and a drone

In this case, we are working with the simplified case of an ACTS composed by a drone, a ground anchor and a point-mass payload. Called with subscript 1 the cable that connects the payload to the drone and with 2 the one that connects it to the ground, for definition the vector u_i, $i = 1 \div n$ is given by

$$u_i = \frac{a_i - p}{\|a_i - p\|} \tag{4.9}$$

being a_i and p respectively the actuator and the payload position.

Payload position control

The goal is to design $F_{prop,1}$ such that the payload reaches a desired position ($p \rightarrow p^*$), which is reachable for the length of the cable.

The payload dynamics is given by:

$$\begin{aligned} m_p \ddot{p} &= \tau_1 u_1 + \tau_2 u_2 - m_p g e_3 \\ &= F_{a,1} + F_{a,2} + F_g \qquad \rightarrow \qquad F_p = m_p(\ddot{p} + g e_3) \\ &= F_p + F_g \end{aligned} \tag{4.10}$$

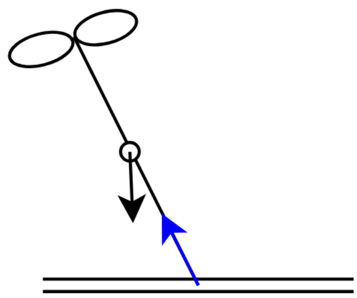


Figure 4.2: Simplified ACTS with one drone and one ground anchor

The desired payload force is defined using an impedance controller:

$$F_p^* = m_p(\ddot{p}^* + ge_3) + K_p(p^* - p) + K_d(\dot{p}^* - \dot{p}) \tag{4.11}$$

with $\dot{p}^* = 0$ and $\ddot{p} = 0$ as we required the payload to maintain the position once it is reached. The remaining force to be generated by cable 1 is:

$$F_{a,1}^* = F_p^* - F_{a,2} \tag{4.12}$$

where $F_{a,2} = \tau_2 u_2$. This brings us back to the situation in Section 4.1.1, where $F_{a,1}$ is f^*.

Tension and orientation control

The goal can also be design $F_{prop,1}$ such that an arbitrary force f^* on the ground can be applied ($F_2 \rightarrow f^*$). In other words, $u_2 \rightarrow u_2^*$ and $\tau_2 \rightarrow \tau_2^*$. This case can be interesting to study as in a redundancy system situation, it may be helpful to add additional constraint in one of the cables. The only difference from the previous case is that the desired position for the payload p^* is not an input anymore, but obtained such that the 2nd cable is aligned along the desired direction given by f^* (Equation 4.2 and 4.3). Then:

$$p^* = a_2 - l_2 u_2^* \tag{4.13}$$

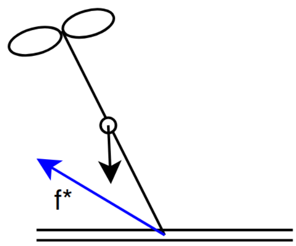


Figure 4.3: Tension directed arbitrary

4.1.3 Case: payload connected to the ground through two cables

The ACTS consists of a point-mass payload connected to the ground through two cables and to an UAV. The enumeration is shown in Figure 4.4.

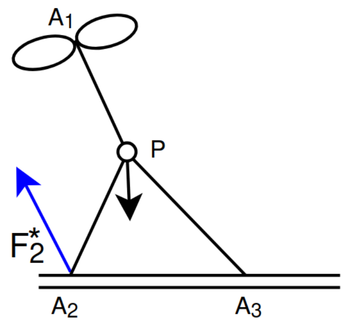


Figure 4.4: Payload connected through two cable to the ground

Payload position control

If the input is the desired payload position p^*, we find ourselves in three situations:

1. p^* lies precisely on the intersecting boundary arcs dictated by the maximum lengths of the ground cables;
2. p^* is out of reach;
3. p^* is inside the triangle A_1A_3P, resulting into slack cables

In the first case, p^* is used to compute the desired payload force F_p^* (Equation 4.11). Using the simplified payload balance equation related to the forces acting on the payload (Equation 3.20), we obtain the following.

$$F_{a,1}^* = F_p^* - \sum_{i=2}^{n} F_{a,i} \tag{4.14}$$

which brings us back to the previous case.

Because the payload is physically constrained at a boundary where both ground cables are stretched straight, any movement or external disturbance is perfectly resisted by the tension of the cables.

In the second case, since the ground cables physically block the payload from reaching p^*, a steady error $e = (p^* - p) \neq 0$ builds up. The actual tension will be directly proportional to the control gains and the distance of the unreachable input:

$$\sum F_{virtual} \propto (p^* - p_{geom}) \quad \Longrightarrow \quad \sum F_{virtual} = K(p^* - p_{geom}) \tag{4.15}$$

where p_{geom} is the closest reachable point that lies on the workspace boundary. In this case, we may impose a controlled tension on the ground cables directed along $(p^* - p_{geom})$ to ensure the drone continues to pull toward the desired target:

$$F_{a,1}^* = F_p^* - \sum_{i=2}^{n} F_{a,i} + F_{virtual} \tag{4.16}$$

In the last case, the distance from the anchors to the payload is less than the cable length, resulting into slack cables as the latter cannot push. Mathematically $\tau_2 = 0$ and/or $\tau_3 = 0$. Then, $F_{a,1}^* = F_p^*$

Desired tension in the cable on the ground

The desired variable to control is the tension of one of the cables $F_{a,2} \to f^*$.

$$F^*_{a,1} = F_p - f^* - F_{a,2} \tag{4.17}$$

where $F_p = m_p(\ddot{p} + ge_3)$ and $F_{a,2} = \tau_2 u_2$.

Desired tension difference between the cables on the ground

The desired variable to control a linear combination of the two $F_{a,2} + \alpha F_{a,3} \to f^*$.

$$F_{a,1} = F_p - F_{a,2} - F_{a,3} \qquad \Longrightarrow \qquad \begin{aligned} F^*_{a,1} &= F_p - (f^* - \alpha F^*_{a,3} + F^*_{a,3}) \\ &= F_p - f^* - (1-\alpha)F_{a,3} \end{aligned} \tag{4.18}$$

For the specific case where $F_{a,2} - F_{a,3} \to f^*$:

$$F^*_{a,1} = F_p - f^* - 2F^*_{a,3} = F_p - f'^* \tag{4.19}$$

4.2 Extension to the Complete ACTS Architecture

At high level, the architecture that enables the system to reach the goal is shown in Figure 4.5.

Trajectory planner

The trajectory planner is able to compute the payload trajectory $x(t)$ and its velocity $\dot{x}(t)$ and acceleration $\ddot{x}(t)$ given the system's goal, such as:

1. Make the payload fly;
2. Bring the payload at the right height and orientation;
3. Approach the façade.

And/or the initial and final position of the payload.

Payload Controller

Then, the desired payload wrench W^*_p is computed for each point in the trajectory based on the previous state of the system. The desired payload wrench is computed from a geometric PD controller. The translational component is given by

$$F^*_p = m_p g + K_p(p^* - p) + K_d(\dot{p}^* - \dot{p}), \tag{4.20}$$

where p^* and p are the desired and current payload positions, respectively. For the rotational dynamics, the controller follows the geometric formulation on $SE(3)$ [20]. The attitude error and angular velocity error are defined as

$$\begin{aligned} e_R &= \frac{1}{2}\left(R^{*\top}R - R^\top R^*\right)^\vee \\ e_\Omega &= \Omega - R^\top R^* \Omega^* \end{aligned} \tag{4.21}$$

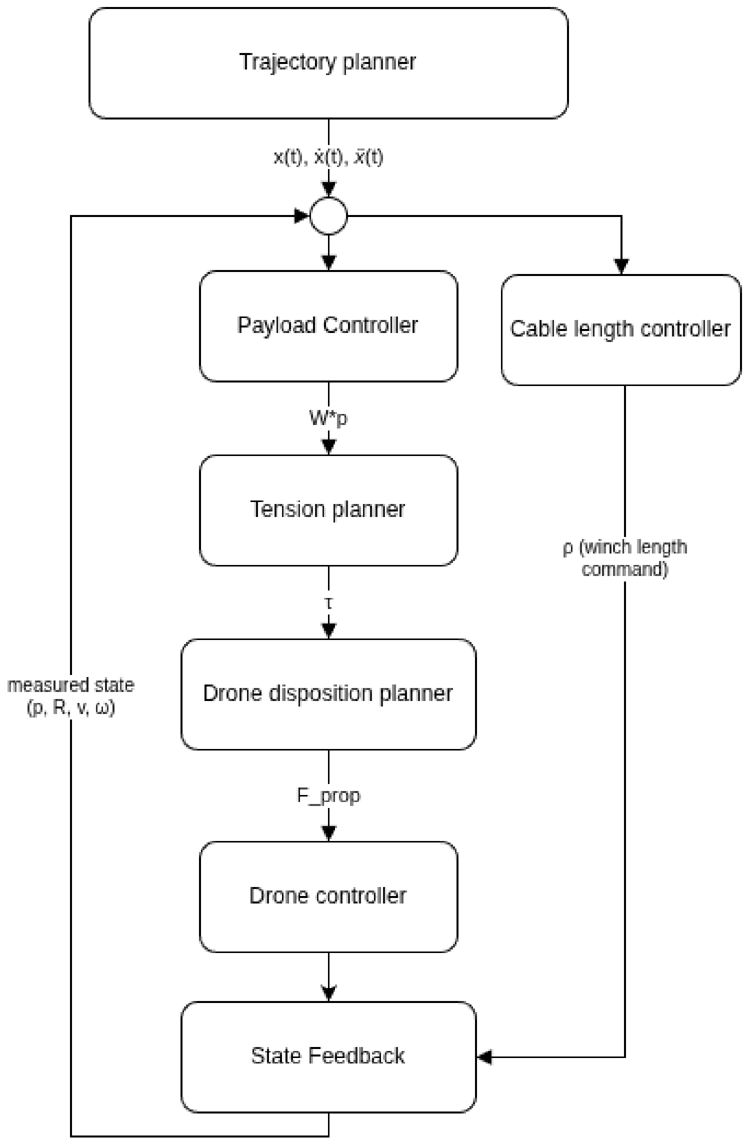


Figure 4.5: High level control scheme

Where $(\cdot)^\vee$ denotes the inverse of the skew-symmetric (hat) operator. Since the controller performs set point regulation, the desired angular velocity is zero: $\Omega_d = \mathbf{0}$, yielding $e_\Omega = \Omega$. The desired control moment in the world frame is then

$$M_p^* = R(-K_R e_R - K_\Omega e_\Omega) \tag{4.22}$$

and the desired payload wrench becomes

$$W_p^* = \begin{bmatrix} F_p^* \\ M_p^* \end{bmatrix}. \tag{4.23}$$

Winch-Level Cable Length Controller

When transitioning the payload to a target pose along the planned trajectory, operational priority is given to the ground actuators due to their significantly higher mechanical stiffness and positioning precision compared to UAVs.

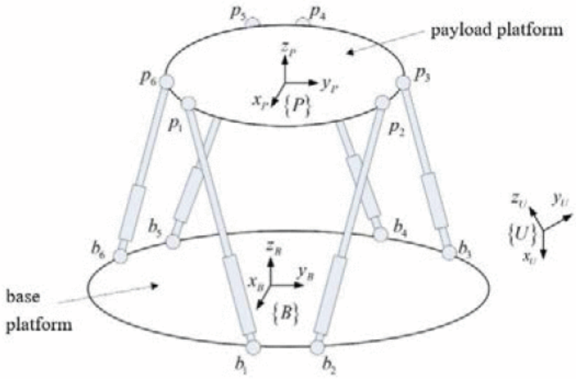


Figure 4.6: Gogh Stewart schematics [8]

The ground subsystem is modeled as a 6-DOF Gough-Stewart parallel mechanism where the six ground cables act as active legs. Given the target payload position p^* and target orientation matrix R^*, the ideal length ρ_i^* for the i-th ground cable is defined geometrically by:

$$\rho_i^* = \|a_i - (p^* + R^* \, {}^p b_i)\| \tag{4.24}$$

To account for the finite velocity of the winch motors, the desired cable length is not applied instantaneously. Instead, the cable-length reference is updated incrementally at each control step according to

$$\rho_i(t + \Delta t) = \rho_i(t) + \Delta\rho_i, \tag{4.25}$$

where the increment $\Delta\rho_i$ is limited such that $|\Delta\rho_i| = v_{\max}\Delta t$, with $v_{\max}$ being the maximum winch speed. The resulting cable-length reference is then sent to the actuator position controller. Consequently, the commanded cable length converges toward the desired value while respecting the physical velocity limits of the winch motors and avoiding unrealistically fast spool motions.

Tension planner

The tension planner is the layer responsible for guaranteeing that every cable stays taut throughout the motion. In fact, when working with ACTS, a payload wrench configuration is statically admissible if and only if there exists a set of strictly positive cable tensions ($\tau_i \geq \tau_{min} > 0$) that balance the target operational wrench. This is due to the fact that cables can only pull and not push. Mathematically, this layout optimization is formulated as:

$$\begin{aligned} \min_{p_a} \quad & \sum_{i=1}^{n_a} \|F_{prop}(p_{a,i}, \tau_{optimal,i})\|^2 + K\|J_p \tau_{optimal} - W_p^*\|^2 \\ \text{subj. to} \quad & x_i^2 + y_i^2 + z_i^2 = l_i^2 \quad \forall i \in [1, n_a] \\ & \rho_{WCC}(\mathcal{U}) \leq 0 \\ & \rho_{IFC}(\mathcal{U}) \leq 0 \end{aligned} \tag{4.26}$$

Which is a Nonlinear optimization Problem (NLP) due to the Euclidean distance constraints.

This objective function aims to minimize the propulsion force of the drones while satisfying the wrench equilibrium equations on the payload. The parameter K acts as a penalty weight that prioritizes the tracking performance over energy efficiency; setting a high value (e.g., $K = 100$), heavily penalizes any mismatch between the generated wrench and the commanded target wrench W_p^*.

For whatever positions p_a, the cable tensions τ are determined by a second inner loop

$$\begin{aligned} \tau_{optimal} = \arg\min_{\tau} \frac{1}{2}\tau^\top H\tau \\ \text{subj. to} \quad J_p^\top(\mathbf{u})\tau = W_p^* \\ \tau_{min,i} \le \tau_i \le \tau_{max,i} \quad \forall i \in [1, n_a + n_g] \end{aligned} \tag{4.27}$$

The lower bound $\tau_{min,i} > 0$ in the wrench feasible condition is what enforces the taut-cable requirement.

As the winch-level control drives $\rho \to \rho^*$, tracking the payload's motion toward its target pose, and $\mathbf{u} \to \mathbf{u}^*$, the achievable wrench set used by the tension planner is itself changing over the trajectory, since it depends on $\mathbf{u}$. As a consequence, $\mathbf{W}_p^*$ may be reachable at the target configuration but momentarily unreachable at the actual configuration, with all tensions kept strictly positive. When this occurs, Equation 4.27 should relax the equality constraint $\mathbf{J}_p^\top(\mathbf{u})\,\tau = \mathbf{W}_p^*$ to a least-squares objective, while keeping $\tau_{min,i} > 0$ as a hard constraint.

$$\begin{aligned} \tau_{optimal} = \arg\min_{\tau} \left(\frac{1}{2}\tau^\top H\tau + \gamma\|J_p^\top(\mathbf{u})\tau - W_p^*\|^2\right) \\ \text{subj. to} \quad \tau_{min,i} \le \tau_i \le t_{max,i} \quad \forall i \in [1, n_a + n_g] \end{aligned} \tag{4.28}$$

This guarantees that the tension planner always returns a taut solution, one that approximates the desired wrench as closely as the actual geometry allows, rather than returning no solution at all during the transient.

For the initial stage of development, this general QP is replaced by a simplified strategy: every cable tension is fixed at the minimum admissible value,

$$\|\tau_i^*\| = \tau_{min}, \qquad \forall i \in [1, n_a + n_g]. \tag{4.29}$$

Drone disposition planner

Once the tension planner computes the scalar target tension magnitudes, the drones must dynamically adjust their orientation and propeller thrusts to impose these forces.

From the static equilibrium balance (Equation 3.17) the required propeller thrust vector is given by:

$$\mathbf{F}_{prop,i} = \mathbf{F}_{a,i} - \mathbf{F}_{g,i} = \mathbf{J}_{a,i}^\top\tau_i + m_i\mathbf{g} = T_i\,{}^0_i\mathbf{R}\,\mathbf{e}_3, \qquad T_i = \|\mathbf{F}_{prop,i}\| \tag{4.30}$$

which is then handed to the attitude controller, responsible for aligning the body z-axis with the commanded thrust vector.

Chapter 5

Simulation Environment and Experimental Setup

5.1 Role of Simulation in the Thesis

The development of the ACTS requires the interaction of several coupled mechanical and control components. A physics-based simulation environment is therefore used to evaluate the proposed architectures and control strategies before physical implementation. Simulation provides a controlled environment in which the system can be tested under repeatable conditions. It also enables the evaluation of the controller without the risks and practical constraints associated with experiments on a physical system.

An additional advantage of using a physics engine is that the complete coupled equations of motion do not need to be explicitly derived and numerically integrated for each configuration. Instead, the mechanical bodies, joints, constraints, actuators, and contact properties are defined in the simulation model, while the physics engine performs the corresponding numerical integration. Nevertheless, the validity of the simulation remains dependent on the assumptions and simplifications made in the model.

5.2 Simulator Selection

Two simulation environments were analyzed during the development of the thesis: Gazebo and MuJoCo. Gazebo was initially considered because of its integration with ROS2 and the availability of existing UAV models and visualization tools. However, difficulties were encountered when attempting to represent the closed cable-constrained structure of the ACTS. MuJoCo was therefore selected for the final simulation environment.

5.2.1 Gazebo Environment

Gazebo Harmonic [21] was initially selected as a potential simulation environment because of its established use in robotics and its integration with ROS2. However, the ACTS architecture presents a particular modeling challenge. Indeed, Gazebo assigns each physical element in the simulation a parent, generating a tree-like structure. While this does not constitute a problem for conventional articulated robot, ACTS contains ground cables constrained at both ends (resp. on the payload and on a point fixed on the ground), resulting in closed kinematic constraints. Additionally, cables as soft body cannot be modelled in Gazebo, as it only manages rigid object. To go around the problem, they are modelled using multiple rigid links connected using universal joints. Although it solved the problem, implementing it resulted into an impracticable slow-down simulation.

Several preliminary simulations were nevertheless successfully implemented in Gazebo, including a pulley mechanism, a UAV connected to the ground through a cable, and a UAV connected to a payload. While individual components of the system could be simulated, such as pulley mechanism and single drone-cable simulation, extending this approach to the complete ACTS resulted in a more complex and computationally demanding model.

These experiments were useful for validating the initial modeling approach and identifying the main limitations before transitioning to another simulation environment. Given the additional modeling effort required to represent the complete ACTS architecture and the limited time available for the thesis, continuing the Gazebo implementation was not considered the most effective approach for obtaining the required simulation results.

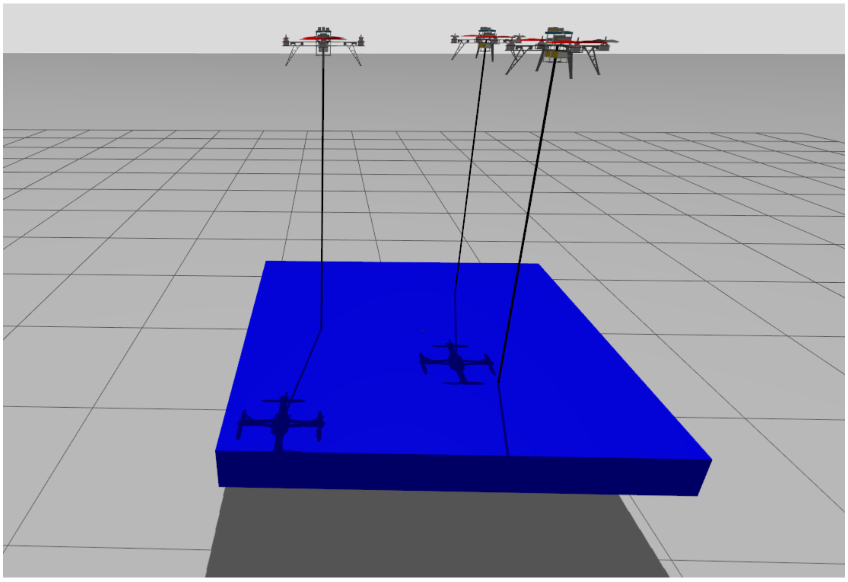


Figure 5.1: Gazebo simulation

5.2.2 MuJoCo Environment

MuJoCo (Multi-Joint dynamics with Contact) was selected as the final simulation environment because it is well suited to the closed-loop structure of cable-driven robots. In particular, MuJoCo provides tendon elements that can be used to represent cable-like connections between bodies, including constraints associated with their geometric configuration [22]. This capability significantly simplifies the representation of the ACTS compared with the rigid-link approximation investigated in Gazebo. The complete system can therefore be represented using the payload, UAVs, ground anchors, and cables while maintaining the closed-loop constraints required by the architecture.

For the simulations presented in this thesis, the cables are modeled as ideal geometric constraints. Their stiffness and damping are neglected, such that the cables transmit tensile forces without introducing additional elastic dynamics. This simplification is consistent with the quasi-static and geometric analysis used for the architecture evaluation, while allowing the simulation to focus on the influence of cable configuration, attachment geometry, and actuator placement.

5.3 Software architecture (UML)

All implementation code[1] is available on GitHub [23]. The repository is organized into three functional groupings:

- a `create_xml` package, an offline configuration and model-generation stage;
- an `acts_simulator` package, a runtime simulation stage;
- a `simulation_run` package, a mass-evaluation stage.

The `create_xml` package handles architecture search and MuJoCo model generation offline, producing XML files that are loaded at startup.
A visual interface (`xml_generator.py`) is also provided to build a custom XML model interactively, ready to be tested without going through the automated architecture search. The resulting models can be run by two independent codes: `acts.py`, a single-model, single-pose driver used for qualitative verification; and `batch_run.py`, a headless driver that iterates over every candidate XML model and every target pose in a batch pose list.

Both codes are meaningful for different reasons. `acts.py` runs one simulation at a time with a live visual interface; this makes it slower, but well suited to the qualitative verification of the controller, which is what it was used for in the early stages of development. Each run saves a single indices row to `collected_data`, together with a video recording and live plots of the tracking performance. `batch_run.py` executes the same underlying pipeline without the visual interface, which makes it considerably faster over many configurations: it accumulates one results row per (model, pose) combination directly into `results.xlsx`. This is the driver used to collect performance data across architectures and poses, from which the conclusions of Chapter 6 are drawn. This overall process is summarized in Figure 5.2.

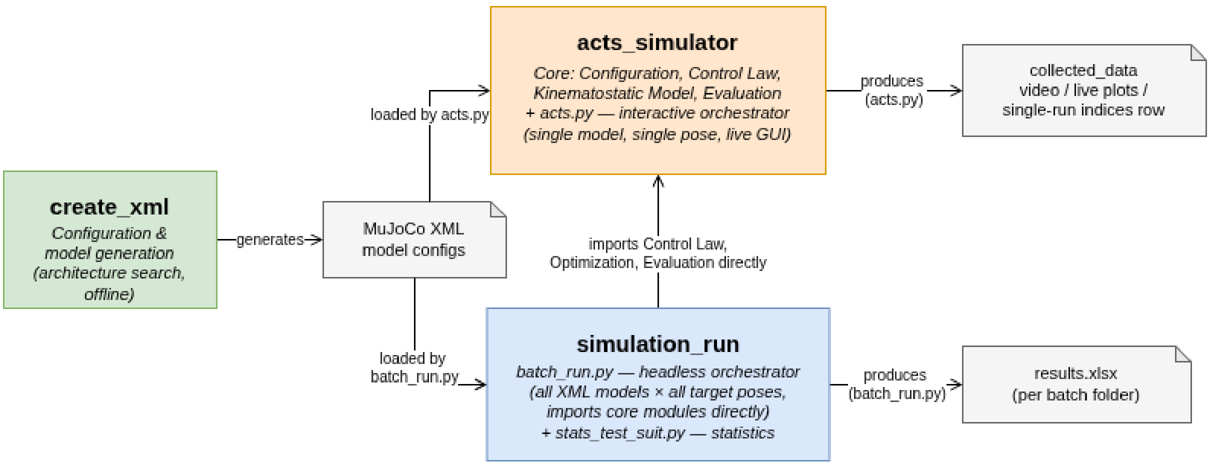


Figure 5.2: Architecture overview

Figure 5.3 details the internal organization of `acts_simulator`. `acts.py` runs the simulation, calling into three modules at each control step:

- the control-law layer (`utils_control.py`), which is described in Figure 5.4

[1] `https://github.com/chiarabuono/acts_simulator.git`

- the kinematostatic/optimization layer (`utils_optimization.py`), which solves for the drone positions and cable tensions that best realize the desired payload wrench
- and the evaluation layer (`utils_performance_indices.py`), which compute the performance indices described in Section 3.3.2 and saves them to file.

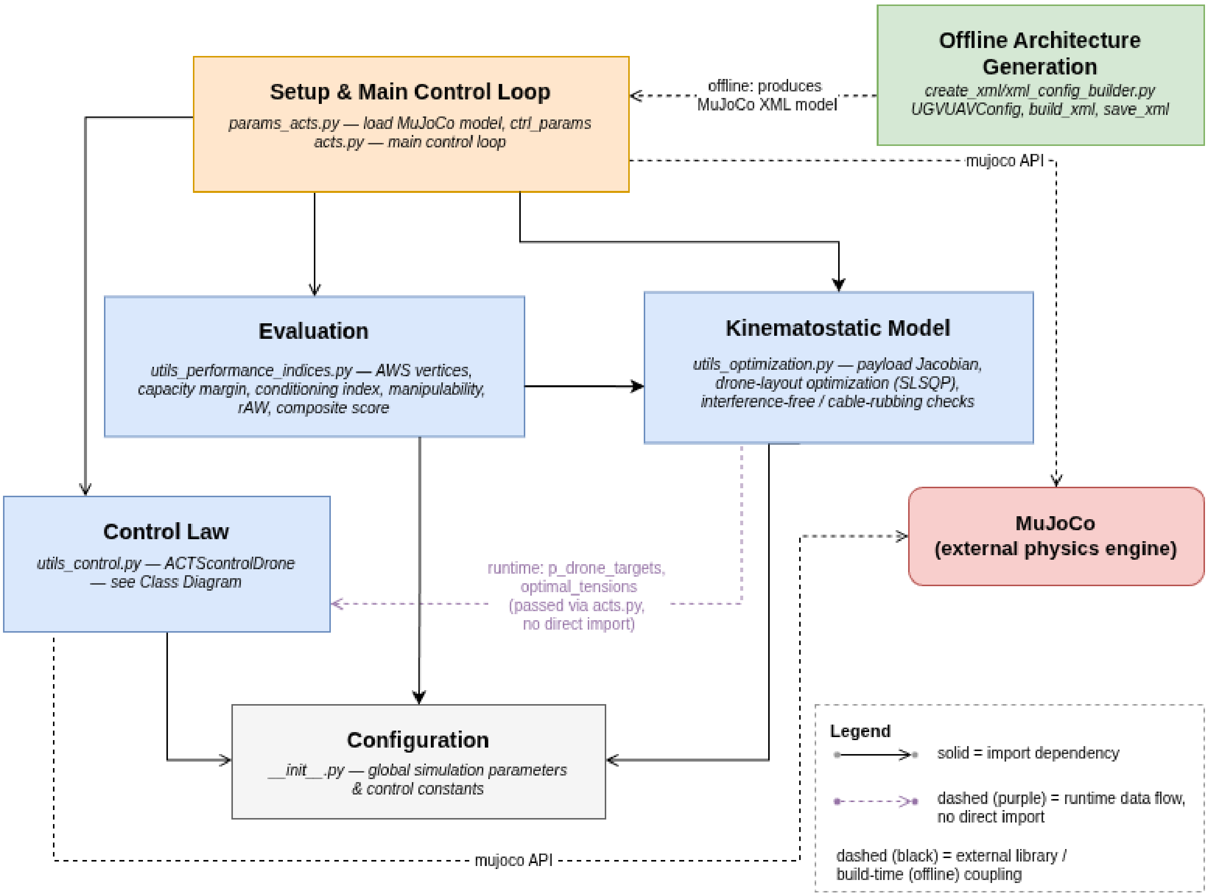


Figure 5.3: Package Diagram

The drone control-law layer introduced above is implemented as a small class hierarchy, as shown in Figure 5.4. In fact, a shared base class, `BaseDrone`, computes everything common to every configuration, such as the motor-mixing matrices, the attitude controller and the necessary rotor speeds. Each subclass replaces only `position_controller()`, providing the force law appropriate to what that drone is physically attached to.

The four subclasses correspond directly to the successive force-control cases developed in Chapter 4, and to the order in which the controller was actually developed and tested:

- `FreeFlightDrone` implements the unconstrained case and was the first to be tested, to verify that a single drone could reach a desired position with no cable constraint at all;
- `CableTetheredDrone` extends this to a drone tethered directly to a fixed ground anchor by a cable described in Section 4.1.1
- `PayloadControlDrone` extends the cable case further to include a payload described in Section 4.1.2;

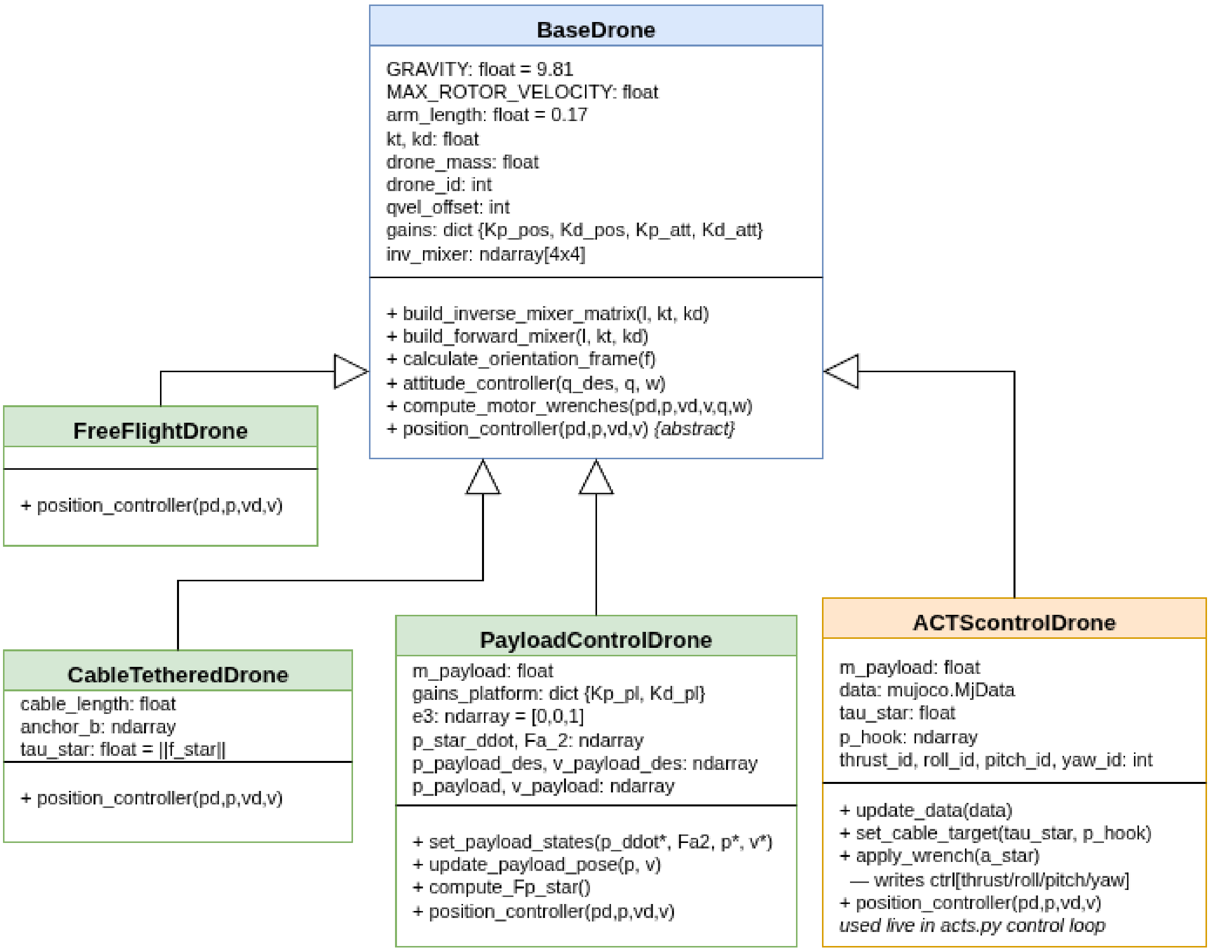


Figure 5.4: Drone Controller Class Diagram

- `ACTScontrolDrone` is the controller ultimately used in the full ACTS architecture, driven directly by `acts.py`.

This progressive structure allowed each force-control case to be validated in isolation before being combined into the final system, and it keeps `position_controller()` as the single, well-defined extension point where a new case could be added without touching the shared attitude and motor-mixing logic.

5.4 Simulation setup

The simulations use the same payload, UAV model, cable properties, controller parameters, and numerical settings to ensure a fair comparison between different architectural configurations. These values are shown in Table 5.1. Additionally, the simulation time is equal to 0.002 seconds. The architectural variables investigated in Chapter 6 are the ground-cable routing, ground-anchor placement, and UAV attachment geometry. At the start of the simulation, the payload is on the ground along with the drones, which are resting on its top surface.

Simulation parameter	Value
Payload and initial conditions	
Payload mass	1 kg
Initial payload position	$[0.0, 0.0, 0.0]^\top$ m
Initial payload orientation	$\mathbb{I}$
Cable and winch parameters	
Minimum cable tension τ_{min}	5.5 N
Maximum cable tension τ_{max}	150 N
Maximum winch speed $v_{\text{winch,max}}$	1.5 m/s
UAV actuation parameters	
Maximum rotor velocity ω_{max}	1666 rad/s
Thrust coefficient k_t	5.5×10^{-6}
Drag coefficient k_d	3.299×10^{-7}
Minimum thrust T_{min}	1.0 N
Maximum thrust T_{max}	$4k_t\omega_{\text{max}}^2$
Safety constraints	
Minimum UAV–UAV distance	0.4 m
Minimum cable–cable distance	0.05 m
Evaluation parameters	
Position tolerance	0.1 m
Orientation tolerance	3°

Table 5.1: Fixed parameters used in the numerical experiments.

5.5 Simulation methodology and evaluation framework

The geometric and static performance indices provided during the model generation stage, describe the instantaneous ability of a given architecture to generate payload wrenches and identify configurations that are close to kinematic singularities. However, these indices are evaluated under static or quasi-static assumptions and therefore cannot fully characterize the behavior of the closed-loop system during motion. In the latter case, cable directions, tensions and payload dynamics continuously evolve. Consequently, a configuration that appears favorable according to static indices may still exhibit oscillations, poor tracking or instability. Closed-loop simulation is therefore necessary to verify whether the theoretical advantages translate into practical performance.

The resulting analysis distinguishes between architectures that are favorable according to the analytical performance indices and those that provide favorable closed-loop behavior in simulation.

Chapter 6

Simulation results

The objective of this chapter is to evaluate how the architectural design of an ACTS influences its ability to accurately and robustly manipulate a payload. The focus is on how different cable routing strategies, UAV attachment configurations, and ground hook placements influence the overall system performance.

In fact, a configuration is considered successful if the desired payload is reached with sufficiently small position and orientation errors, all cables remain under positive tension throughout the maneuver, and the payload converges to an equilibrium without persistent oscillations.

First, the classical Stewart configuration is analyzed and used as a reference architecture due to its geometric symmetry and extensive use in the literature. In fact, if even Stewart-like performs poorly, the problem is likely the controller rather than the architecture.

Two desired payload poses are taken in considerations. In the first scenario, the desired payload orientation coincides with the world reference frame; while in the other scenario a non-zero desired orientation is imposed. This allows to analyze separately the rotational motions from the translation tracking.

The remaining experiments then investigate how modifications to the ground cable routing, UAV attachment geometry, and hook placement affect both the theoretical performance indices and the closed-loop behavior of the system. This progressive analysis makes it possible to identify which geometric features have the greatest influence on payload manipulation performance.

6.1 Stewart platform baseline

The Stewart platform configuration is used as a baseline for evaluating the proposed ACTS architectures. Its symmetric arrangement of the six ground cables provides a well-defined reference configuration against which alternative cable routings can be compared. The baseline is first examined using the static performance indices introduced in Section 3.3.2.

In particular, the conditioning index, manipulability, radius of available wrench, capacity margin, and composite score are evaluated at the prescribed payload pose. These quantities provide a reference for assessing the wrench-generation capability and isotropy of the ground-cable arrangement. The complete ACTS configuration is then evaluated by incorporating the aerial cables and their actuation constraints. Finally, the resulting configuration is simulated in MuJoCo to verify the dynamic payload tracking behavior.

6.1.1 Numerical Verification of the Baseline Configuration

This section verifies the implementation of the optimization and control algorithms by independently reconstructing the principal computation steps for a representative Stewart-platform configuration. The payload Jacobian, cable tensions, payload wrench, and UAV thrust commands are recomputed from the analytical equations derived in Chapter 3 and compared with the implementation. Agreement between the analytical calculations and the numerical results confirms the correctness of the implementation before comparing different ACTS architectures.

Reference configuration

Payload dimensions: $2.0 \times 2.0 \times 0.2$ m
Desired pose: $p^* = [0.5, -0.5, 2.0]^\top$ m
Desired orientation: $R^* = \mathbb{I}$
Desired payload wrench: $W_p^* = [0, 0, 9.81, 0, 0, 0]^\top$

attachment		hook points	
$A_4 \equiv A_5$	$[0.0, 1.1, 0.0]^\top$	$B_4 \equiv B_9$	$[0.5, 0.05, 1.9]^\top$
$A_6 \equiv A_7$	$[-0.9526, -0.55, 0.0]^\top$	$B_5 \equiv B_6$	$[-0.0237, -0.775, 1.9]^\top$
$A_8 \equiv A_9$	$[0.9526, -0.55, 0.0]^\top$	$B_7 \equiv B_8$	$[0.9763, -0.775, 1.9]^\top$

Table 6.1: Ground anchor and hook attachment points for the Stewart baseline.

Ground System Geometry

Using Equation 3.3b, the ground cable directions u_i and the corresponding Jacobian are obtained as:

$$
\begin{aligned}
u_4 &= [-0.2245,\ 0.4713,\ -0.8529]^\top & u_7 &= [-0.7100,\ 0.0828,\ -0.6993]^\top \\
u_5 &= [-0.0089,\ 0.7024,\ -0.7117]^\top & u_8 &= [-0.0124,\ 0.1176,\ -0.9930]^\top \\
u_6 &= [-0.4545,\ 0.1047,\ -0.8846]^\top & u_9 &= [0.2215,\ -0.2937,\ -0.9299]^\top
\end{aligned}
$$

Table 6.2: Ground cable unit directions.

$$
J_{p,\text{ground}} = \begin{bmatrix}
-0.22445 & 0.47135 & -0.85291 & -0.42197 & 0.02245 & 0.12345 \\
-0.00888 & 0.70238 & -0.71175 & 0.26597 & -0.33812 & 0.33656 \\
-0.45452 & 0.10475 & -0.88456 & 0.25373 & -0.37586 & 0.29632 \\
-0.70998 & 0.08282 & -0.69934 & 0.20060 & 0.40409 & -0.52834 \\
-0.01239 & 0.11759 & -0.99298 & 0.28483 & 0.47420 & -0.47636 \\
0.22151 & -0.29365 & -0.92989 & -0.54081 & -0.02215 & -0.12183
\end{bmatrix} \tag{6.1}
$$

Drone Attachment Geometry

The UAV positions are not predefined but result from the tension-planning procedure described in Section 4.2. The corresponding cable directions u_1, u_2, u_3 are therefore extracted from a converged simulation run for this configuration rather than computed from known attachment coordinates:

$$
\begin{aligned}
a_1 &= [1.1020, -0.2692, 3.5972]^\top \\
a_2 &= [0.1802, -0.7553, 3.4976]^\top \\
a_3 &= [0.9397, -1.2760, 3.5192]^\top
\end{aligned}
\tag{6.2}
$$

Using Equation 3.3b, the resulting drone cable directions are

$$
\begin{aligned}
u_1 &= [0.0837, -0.0294, 0.9961]^\top \\
u_2 &= [0.1042, -0.3528, 0.9299]^\top \\
u_3 &= [0.2926, -0.1504, 0.9443]^\top
\end{aligned}
\tag{6.3}
$$

with $\|a_i - r_i\| \approx 1.503$ m for all three, consistent with the nominal drone cable length of 1.5 m and confirming the optimizer converged to a geometrically taut solution. The corresponding drone Jacobian is

$$
J_{p,\text{drones}} = \begin{bmatrix} 0.0837 & -0.0294 & 0.9961 & 0.2769 & -0.4661 & -0.0370 \\ 0.1042 & -0.1042 & 0.9299 & 0.2910 & 0.4533 & 0.1394 \\ 0.2926 & -0.1504 & 0.9443 & -0.5043 & 0.0293 & 0.1609 \end{bmatrix}
\tag{6.4}
$$

Cable Tensions

For this configuration, the optimizer returns

$$
\begin{aligned}
\tau_{\text{UAV}} &= [13.6970,\ 11.1548,\ 14.4154]^\top \text{ N} \\
\tau_{\text{ground}} &= [5.4903,\ 5.4740,\ 5.4901,\ 5.4784,\ 5.4860,\ 5.4708]^\top \text{ N}
\end{aligned}
\tag{6.5}
$$

All six ground tensions lie close to $\tau_{\text{min}} = 5.5$ N. This result is expected. Because the desired wrench is almost entirely vertical, the UAVs provide most of the lifting force while the Stewart cables primarily stabilize the payload orientation and prevent lateral motion. Consequently, the optimizer drives the ground tensions toward their minimum admissible values, maintaining taut cables while minimizing unnecessary internal forces.

Payload Wrench Verification

Because the tension planner solves a bounded least-squares problem, the recovered wrench is not expected to match the desired wrench exactly.

Combining $J_{p,\text{drones}}$ and $J_{p,\text{ground}}$ into the full Jacobian J_p and evaluating $W_p = J_p\,\tau$ with the tensions above:

$$
W_p = J_p\,\tau = \begin{bmatrix} -0.0023 \\ 0.0018 \\ 9.8110 \\ -0.0003 \\ -0.0001 \\ -0.0009 \end{bmatrix}, \qquad W_p^* = \begin{bmatrix} -0.0158 \\ 0.0180 \\ 9.7341 \\ -0.0022 \\ 0.0019 \\ -0.0151 \end{bmatrix}.
\tag{6.6}
$$

All six components match to within 0.08 in force units and 0.015 in moment units. The residual remains small, confirming that the optimizer reproduces the desired wrench within the prescribed tolerance.

Required Drone Thrust

Under the steady-state assumption $p_d = p$, $\dot{p}_d = \dot{p} = 0$, Eq. 4.30 reduces to $F_{prop,i} = \tau_i u_i + m_i g\, e_3$, giving the hand-computed thrust magnitudes

$$T_1 = 33.29 \text{ N}, \qquad T_2 = 30.27 \text{ N}, \qquad T_3 = 33.57 \text{ N}. \tag{6.7}$$

The controller-computed thrust commands for the same converged pose are 33.2793, 30.2683, and 33.5657 N respectively, matching the hand-computed values to within numerical precision and confirming the correct implementation of Eq.4.30.

Discussion

The numerical verification confirms the correctness of both the optimization and control implementations. The recovered payload wrench satisfies the desired equilibrium within the expected residual introduced by the bounded least-squares tension planner, while the hand-computed UAV thrusts coincide with the controller outputs to numerical precision. These results provide confidence that the subsequent comparison of ACTS architectures reflects differences in the robot configurations rather than implementation errors.

6.1.2 Comparison between tension allocation strategies

As mentioned in Section 4.2, we evaluate two distinct tension allocation strategies:

- Minimal tension strategy, in which a minimum cable tension is imposed to each cable tension such that $\|\tau_i\| = \tau_{min}$.
- Optimal tension strategy, in which cable tensions are determined by the quadratic optimization problem in Equation 4.28

The performance indices and tracking errors under both strategies across two operational scenarios are summarized in Table 6.3.

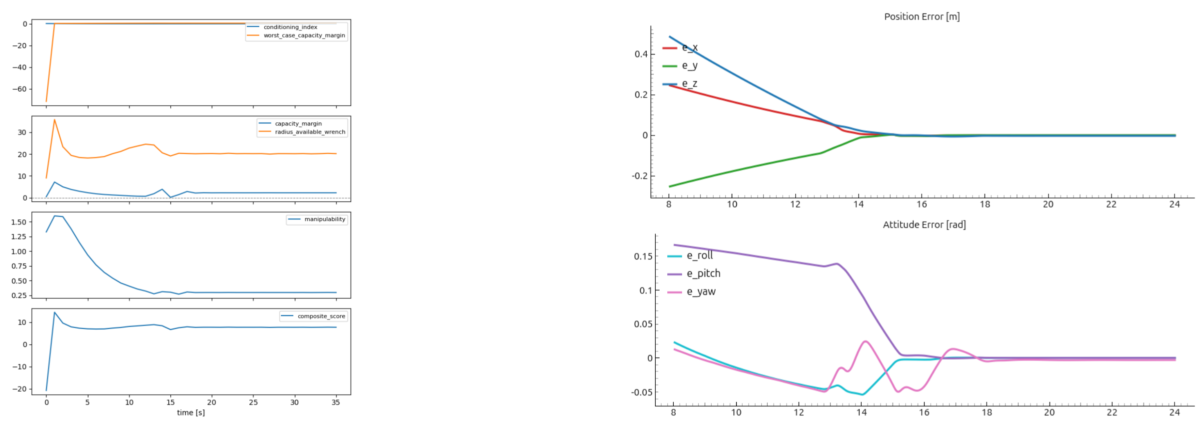
The minimal tension strategy has negative consequences on the system's robustness. By locking all cable tensions to their absolute lowest limit ($\tau_{\min}$), the range of available tension choices shrinks to a single point. As a result, the controller loses its ability to adjust forces, reducing the payload's Available Wrench Set (AWS) to just a single fixed outcome. [24] As a result, any deviation of the desired payload wrench W_p^* from this single point results in a negative capacity margin (as seen in Scenario A where $\gamma = -0.19343$). This indicates that the system is theoretically incapable of resisting external disturbances or maintaining equilibrium without violating the tension constraints.

Switching to the optimal tension allocation, allows the cable tensions to vary dynamically between τ_{min} and τ_{max}. As a result, the radius of the Available Wrench increases and the capacity margin takes on a positive value ($\gamma = 10.46403$). Thus, also the composite score increases. Differently, changing the tension allocation algorithm has no effect on the conditioning index or manipulability because they are kinematic metrics that only depend on the platform's geometric position.

On the other hand, it is worth to notice that, since the objective function does not contain any state-tracking terms, this optimizer does not directly minimize the payload's real-time position or orientation tracking errors. Because of this, a spatial configuration that yields globally minimal propulsion effort does not guarantee a minimum in tracking error.

Metric / Parameter	Scenario A		Scenario B	
	$\tau_{\min}$	τ_{opt}	$\tau_{\min}$	τ_{opt}
Desired Position	$[0.5, -0.5, 2.0]^\top$		$[0.5, -0.5, 2.0]^\top$	
Desired Orientation	$[1.0, 0.0, 0.0, 0.0]^\top$		$[1.0, 0.1, -0.1, 0.1]^\top$	
Conditioning Index	0.10259	0.09617	0.07052	0.05550
Manipulability	0.47775	0.43700	0.31471	0.24108
Radius Avail. Wrench	49.29711	47.90399	52.21278	53.22707
Capacity Margin	-0.19343	10.46403	-3.83256	2.65386
Worst-case CM	0.90998	0.76401	0.90998	0.75072
Composite Score	17.35180	20.62849	17.64446	19.42488
Position Error	0.00201	0.00284	0.00187	0.00262
Orientation Error	0.00121	0.00209	0.00379	0.00385

Table 6.3: Comparative performance indices and tracking errors for Stewart baseline configuration across Scenarios A and B.



(a) Performance indices

(b) Pose error

Figure 6.1: Graphs

Although the Stewart configuration exhibits good overall performance and provides an excellent reference architecture, the proposed ACTS differs from a classical Stewart platform because the UAV cables can be connected to the payload in several different ways. Moreover, different ground cable routings may further improve specific performance aspects such as stability, wrench generation capability, or robustness. For these reasons, the Stewart configuration is adopted as a baseline against which all alternative architectures are compared.

An ideal ACTS architecture should remain far from singular configurations, generate payload wrenches efficiently in every direction, and maintain sufficient redundancy to accommodate disturbances while respecting cable tension limits. Consequently, all the considered performance indices should be as big as possible.

The considerations from now on will be done using the optimal tension strategy.

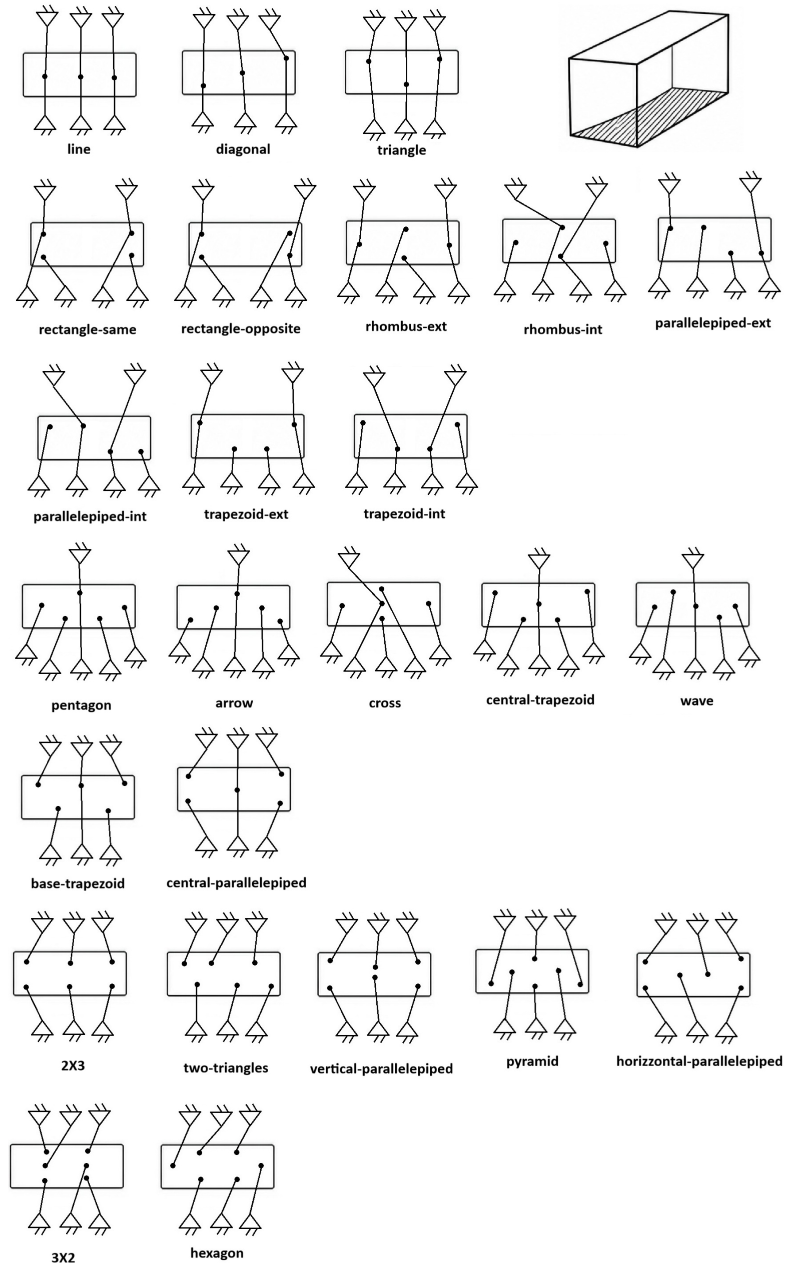


Figure 6.2: Possible 6 UGVs connections

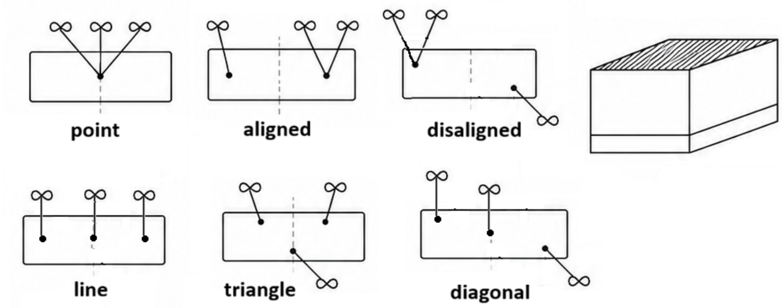


Figure 6.3: Possible 3 UAVs connection with one cable

6.2 Statistical tests

The performance of each robot configuration was evaluated using a set of continuous and binary performance metrics extracted from the simulations. Since every configuration was assessed on the same set of target poses, the resulting observations constitute a repeated-measures design.

Continuous metrics were analysed using the Friedman test, whereas binary metrics were analysed using Cochran's Q test. For metrics exhibiting a statistically significant omnibus result, post-hoc comparisons against the reference configuration were performed using paired Wilcoxon signed-rank tests for continuous variables and McNemar's exact tests for binary variables.

To account for multiple testing, the Benjamini–Hochberg false discovery rate (BH–FDR) procedure was applied to the omnibus tests, while Holm's correction was used for all post-hoc pairwise comparisons. Statistical significance was assessed using a significance level of $\alpha = 0.05$.

A detailed description of the adopted statistical methods is provided in Appendix B.

6.3 Ground cable routing

Once the controller has been validated using the Stewart baseline, we can proceed to consider different ways to connect six ground cables to the platform (Figure 6.2) and the drone (Figure 6.3). The same ground configuration hook-anchor can change the system's stability based on which hook is connected to which anchor point on the ground. Therefore, we will start considering the drone attachment configuration as fixed to the triangular arrangement and vary only the routing of the six ground cables. This specific UAV configuration is chosen because it provides a symmetric distribution of the three drones around the payload. This approach allows one to isolate the influence of the ground cable routing and attribute differences in performance solely to the ground architecture. Finally, in Section 6.5, the impact of various UAV attachment geometries is examined independently.

Focusing on the ground cable routing, there are at most $6! = 720$ physically distinct valid ways to route the six ground cables. Scoring an architecture using only one arbitrarily chosen routing raise the question on how lucky the routing choice was and how good the architecture actually is. Therefore, implementing the selection strategy explained in Section 3.3.3 is essential to make a guided choice and conclusions on the results.

6.3.1 Ground screening

The pre-screening results are shown in Table 6.4 and Table 6.5, where ground cable routings were tested over 10 different poses. Any routing that violates either WCC or IFC at any tested pose is removed, as it was explained deeply in Section 3.3.3. For this reason, the WCC and IFC counts in Table 6.4 are per-routing failures, while the WFC count is not directly comparable, since it represents the total number of pose-level failures accumulated across all candidate routings. The WFC is the main cause of rejection, which means that many routings are still kinematically feasible, but they cannot produce the required wrench while keeping the cable tensions within the allowed range. The WCC failures are relatively few, indicating that rank deficiency is less common than tension infeasibility. IFC failures are only caused by cable-cable interference, and not by exit-angle violations.

Configuration	**WCC Failures**	**WFC Failures**	**IFC Failures**	**Feasible**
pentagon – rectangle-same	0	394	42	✓
two-triangles – 2X3	72	2300	485	✓
two-triangles – rectangle-same	6	787	110	✓
rectangle-opposite – 2X3	72	146	94	✓
parallelepiped-int – 2X3	52	340	102	✓

Table 6.4: Pre-screening feasibility of cable-routing architectures, triangle UAV topology

The quality of the selected routing is evaluated through four complementary metrics. The smallest singular value, $\sigma_{\min}$, measures the distance from singular configurations, while the conditioning index ν evaluates the isotropy of the wrench generation capability. Manipulability quantifies the overall size of the achievable wrench space, whereas $d_{\min}$ characterizes the minimum cable-to-cable clearance over all tested poses.

Configuration	$\sigma_{\min}$ [a]	ν [b]	w [c]	$d_{\min}$ [d]
pentagon – rectangle-same	0.2005	0.0832	0.0721	0.1790
two-triangles – 2X3	0.1965	0.0820	0.0628	0.0638
two-triangles – rectangle-same	0.1893	0.0793	0.0740	0.0982
rectangle-opposite – 2X3	0.1875	0.0786	0.0957	0.2027
parallelepiped-int – 2X3	0.1892	0.0784	0.0344	0.3213

Table 6.5: Best routing quality across feasible architectures

[a] $\sigma_{\min}$ = smallest singular value of the payload Jacobian.
[b] ν = conditioning index (Eq. 3.29)
[c] w = manipulability (Eq. 3.30)
[d] $d_{\min}$ = minimum cable-to-cable clearance of the selected routing

Table 6.5 shows how no single architecture simultaneously maximizes every performance metric. The `pentagon--rectangle-same` configuration provides both the largest minimum singular value and the best conditioning index, indicating the greatest robustness against singular configurations among this candidate set.
In contrast, `rectangle-opposite--2X3` achieves the highest manipulability ($w = 0.0957$). Finally, `parallelepiped-int--2X3` provides by far the largest minimum cable clearance ($d_{\min} = 0.3213$), more than three times that of the best-conditioned routing.

The triangle-triangle pairing is absent from Tables 6.4 and 6.5 because both the payload and the ground layouts contain only three physical attachment locations, with two cables attached at each vertex. Since switching tw cables connected to the same node does not change the routing geometry, only the correspondence between the three payload and three ground attachment clusters must be considered, resulting in just $3! = 6$ distinct routings instead of the 180-720 evaluated for the other layouts.

However, the reduced number of routings does not imply that the architecture is kinematically infeasible. Although three of these six routings satisfy the wrench-closure condition (WCC), none survives the subsequent screening stages. Consequently, the triangle-triangle architecture is rejected because it cannot maintain feasible cable tensions or interference-free operation across the tested workspace, rather than because of the reduced number of candidate routings.

Quantity	Value
Distinct routings evaluated	6
Routings satisfying WCC	3
WCC failures	3
WFC failures (10 poses)	18
IFC failures	1
Feasible routings	0

Table 6.6: Screening outcome for the triangle-triangle architecture.

6.3.2 Simulation run

The theoretical indices evaluate the geometric and static properties of a cable routing. They define configurations close to kinematic or wrench singularities, and they classify architectures based on their ability to generate payload wrenches. However, they cannot predict the transient response of the closed-loop system. Dynamic effects such as payload inertia and load redistribution are only captured by simulation. Therefore, while poor theoretical indices almost always indicate poor dynamic performance, the opposite is not necessarily true: a routing with good static indices may still have oscillations or bad tracking.

For each configuration derived in the previous step, a simulation is run over 60 different poses (shown in Figure 6.4), and conclusions are drawn. Additionally, extra simulations have been run to test specific ground cable routing and see if they lead us to further conclusions. Table 6.7 summarizes the results of the dynamic simulations for the candidate architectures selected during the pre-screening. The objective is to evaluate whether the theoretical metrics introduced previously are consistent with the observed closed-loop behavior. In particular, the simulations assess pose reachability, cable rubbing and numerical stability over the set of 60 testing poses. Each indicator is ranked from 1 to 5, where lower values correspond to better overall performance.

Architecture	ν	w	N	Pose Reach	Cable Rub	Instab.
rectangle-opposite-2x3-triangle	0.1246 (1)	1.0459 (1)	8.9142 (1)	0.5833 (6)	0.0000 (1)	0.0500 (2)
two-triangles-rectangle-same-triangle	0.1052 (3)	0.4603 (3)	7.3013 (2)	0.6500 (4)	1.0000 (6)	0.1833 (6)
parallel-int-2x3-triangle	0.0997 (4)	0.4536 (4)	6.7857 (3)	0.7500 (2)	0.0000 (1)	0.0500 (2)
pentagon-rectangle-same-triangle	0.0869 (5)	0.3522 (5)	6.4366 (4)	0.8333 (1)	0.6500 (4)	0.0000 (1)
two-triangles-2x3-triangle	0.1140 (2)	0.5953 (2)	6.2580 (5)	0.6667 (3)	0.7667 (5)	0.0833 (4)
acts_stewart	0.0533 (6)	0.1054 (6)	5.5706 (6)	0.6500 (4)	0.3167 (3)	0.0833 (4)

Table 6.7: Static and dynamic performance indices for the candidate architectures. Rankings (1 = best) are reported separately for each metric.

ν = conditioning index; w = manipulability; N = composite score.
Pose Reach = fraction of poses successfully reached.
Cable Rub = fraction of simulations with cable rubbing (lower is better).
Instab. = fraction of unstable simulations.

The simulation results only partially agree with the theoretical ranking obtained during the pre-screening stage. The `rectangle-opposite-2x3-triangle` architecture

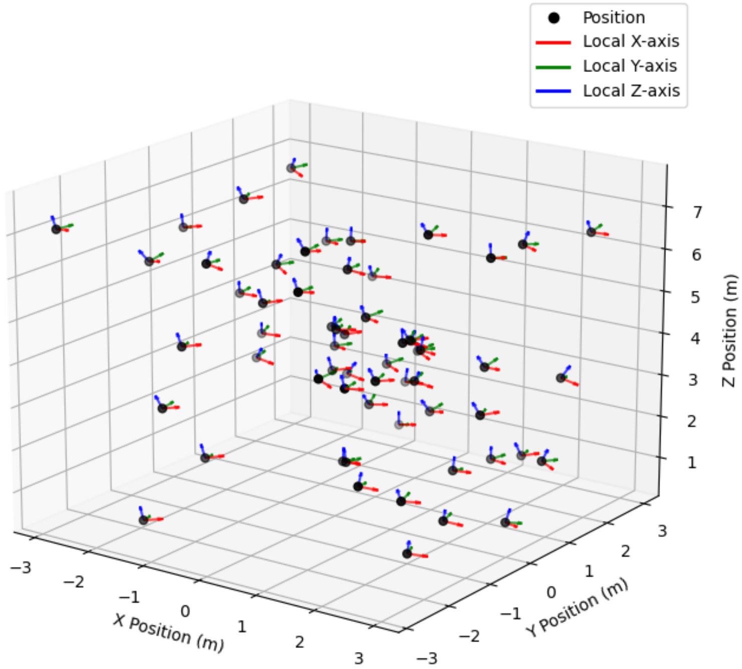


Figure 6.4: Testing poses

achieves the highest conditioning index, manipulability and composite score, confirming its excellent geometric properties. However, it also exhibits the lowest pose reachability, indicating that favorable static indices alone do not guarantee successful closed-loop operation. Conversely, the `pentagon-rectangle-same-triangle` architecture, despite ranking only fourth according to the theoretical metrics, achieves the highest pose reachability and is the only architecture with no unstable simulations. Similarly, the `parallelepiped-int-2x3-triangle` architecture combines good pose reachability with zero cable rubbing, making it one of the best overall dynamic performers despite not being the top-ranked architecture according to the static indices.

These observations demonstrate that the theoretical indices are effective for identifying architectures with good geometric properties and excluding clearly unsuitable routings, but they are insufficient to predict the dynamic behavior of the complete system. Dynamic phenomena such as payload inertia, cable tension redistribution and the interaction between the controller and the mechanical structure strongly influence the final performance and therefore require time-domain simulation.

The comparison between the pre-screening metrics and the dynamic simulations highlights the complementary role of the two evaluation stages. The theoretical metrics successfully identify architectures with favorable wrench-generation capabilities and good conditioning, making them an effective tool for reducing the number of candidate routings before simulation. Nevertheless, the dynamic simulations reveal that these metrics do not directly translate into superior closed-loop performance.

For example, the rectangle-opposite-2x3-triangle architecture achieves the best conditioning index, manipulability and composite score, yet it exhibits the lowest pose reachability. In contrast, the pentagon-rectangle-same-triangle architecture, which ranks only fourth according to the theoretical indices, achieves the highest pose reachability while remaining numerically stable throughout all simulations. Likewise, the parallelepiped-int-2x3-triangle architecture combines high pose reachability with zero cable rubbing, illustrating that different routing strategies may outperform the theoretically optimal solution once the system dynamics are considered.

Supplementary simulations have been run on configurations realized manually, rather than selected through a systematic search. They demonstrate that the cable routing itself has a significant influence on the dynamic behavior, even when the underlying payload and ground geometries remain unchanged. Architectures sharing the same layouts but employing different cable assignments exhibit noticeable differences in pose reachability, cable rubbing and instability. Overall, the manually selected routings do not consistently outperform those obtained through the systematic screening procedure. This suggests that manually identifying a suitable routing based solely on geometric intuition is difficult, whereas the proposed screening strategy provides a more reliable and reproducible means of selecting high-quality candidate routings for subsequent dynamic evaluation.

Architecture	ν	w	N	**Pose Reach**	**Cable Rub**	**Instab.**
rect-same–rect-opposite	0.1077 (2)	9.1356 (1)	9.1356 (1)	0.6167 (5)	1.0000 (4)	0.2500 (3)
2X3–rect-same	0.1226 (1)	8.5115 (2)	8.5115 (2)	0.7000 (3)	1.0000 (4)	0.4833 (5)
horiz-parall.–horiz-parall.-3	0.0647 (3)	6.6252 (3)	6.6252 (3)	0.7833 (1)	0.4000 (1)	0.0667 (1)
horiz-parall.–horiz-parall.	0.0645 (4)	6.5651 (4)	6.5651 (4)	0.7167 (2)	1.0000 (4)	0.2500 (3)
rect-opposite–2X3-2	0.1126 (5)	6.4080 (5)	6.4080 (5)	0.5667 (4)	1.0000 (4)	0.1667 (2)

Table 6.8: Top five manual architectures

6.3.3 Analysis of Specific configurations

Degenerative ground configurations - Line and Diagonal

The exhaustive screening consistently rejects the line and diagonal ground-routing configurations. This result is expected, as in these layouts several cable directions become nearly coplanar or aligned, resulting in cable arrangements that are close to kinematic singularities. Consequently, the cable system can no longer generate a full six-dimensional wrench space. This result could have also been explained geometrically. In the line configuration, cable attachments lie along a single axis, which limits the system's ability to generate forces and torques in directions perpendicular to that line. Similarly, the diagonal configuration introduces a strong directional bias along the axis on which the cable attachments are positioned.

Simulation results using hand-crafted configurations confirm this conclusion: line-based and diagonal-based ground cable layouts never rank among the highest-performing architectures in terms of composite score. The drop in performance becomes particularly pronounced when both the payload and the ground attachments are arranged in a one-dimensional configuration. In such cases, the desired pose is achieved less than half the time, like in the line-line-triangle configuration that succeeds only 40% of the time. Success rates nearly double when at least one attachment expands into two dimensions, as seen in line-pentagon-triangle (56.67%) or line-hexagon-triangle (76.67%).

6.4 Ground anchor placement

The influence of the ground-anchor spacing was evaluated by comparing three geometrically similar datasets in which the ground attachment locations were uniformly scaled while preserving the same cable-routing topologies. The comparison was restricted to the triangle-UAV configurations to isolate the effect of anchor placement from that of cable routing.

Increasing the distance between the ground anchors is expected to improve the robot's mechanical performance. A larger anchor footprint increases the moment arm between the payload and the cable attachment points, allowing the cables to generate larger restoring moments and a wider achievable wrench set. Consequently, higher wrench quality, better conditioning, and improved tracking performance are expected.

The complete numerical results for all evaluated metrics, workspace sizes, and platform configurations are reported in Appendix C. Tables C.1 and C.2 provide the complete metric values together with the statistical significance of each comparison relative to the reference `acts_stewart` configuration.

6.4.1 Results with reduced anchor spread

Reducing the anchor spacing consistently degrades the performance of all alternative architectures. Static performance indicators (Table C.1), including the capacity margin, manipulability, radius of available force, and conditioning index, decrease across almost every configuration. Likewise, cable rubbing becomes more frequent as the anchors are brought closer together, indicating that the cable directions become increasingly parallel and clearance between neighbouring cables is reduced.

The effect is particularly evident in the task-level metrics (Table C.2). The pose reached metric shows the clearest deterioration: whereas the reference Stewart configuration successfully reaches 65% of the target poses, all compressed alternatives achieve only 43-65%. The omnibus Cochran's Q test indicates a statistically significant difference in pose reachability among the evaluated configurations ($p = 0.03256$) and is reflected in the number of unreached poses, which increases from 5/60 in the nominal dataset to 10/60 in the reduced-scale dataset. However, post-hoc McNemar tests with Holm correction did not identify any individual configuration as significantly different from the reference Stewart architecture.

6.4.2 Results with increased anchor spread

Increasing the anchor spacing produces the opposite trend. Every static performance metric improves, with higher capacity margins, larger radii of available force, better conditioning indices, and increased manipulability across all evaluated architectures. Position and orientation tracking errors also decrease for nearly every routing, demonstrating that the improved static properties translate into better closed-loop performance.

The wider anchor distribution also improves task completion. All target poses are reachable, compared with five in the nominal configuration and ten in the reduced-scale configuration (Table 6.9). Interestingly, the overall difference in pose completion between routing topologies is no longer statistically significant ($p = 0.07573$). This suggests that, once the anchor spread is sufficiently large, the system operates well within its feasible wrench region and the choice of cable routing becomes considerably less critical.

6.4.3 Considerations on the anchor spacing

The experimental results consistently support selecting the largest feasible ground-anchor footprint. Compared with the nominal and reduced configurations, the larger footprint provides superior composite performance, larger wrench margins, better conditioning, lower tracking errors, and substantially fewer unreached poses. Furthermore, it is the only tested scale for which the routing topology no longer has a statistically significant influence on pose completion, indicating increased robustness to architectural choices.

The improvement in reachability is also spatially meaningful. As the anchor spacing increases, failures at low altitude disappear completely, while the few remaining unreached poses are located near the upper boundary of the tested workspace. This behavior is consistent with the underlying mechanics: increasing the anchor spacing enlarges the available moment arm most effectively when the payload is close to the ground anchors. At greater heights, the cable directions become nearly vertical regardless of the anchor spacing, reducing the benefit of further increasing the footprint.

The principal drawback of a larger anchor spacing is practical rather than mechanical. A wider anchor distribution requires a larger installation area around the deployment site, which may be constrained by the available space during façade operations. Consequently, although largest one provides the best mechanical performance, the final design should balance these gains against the physical limitations of the intended deployment environment.

Ground-anchor footprint	Unreached poses
Small	10/60
Nominal	5/60
Large	0/60

Table 6.9: Number of target poses that remained unreachable for each evaluated ground-anchor spacing.

6.5 Drone hook placement

This section focuses on analyzing how the UAVs attach to the payload. First, we compare a common cable-routing scheme across all configurations shown in Figure 6.3, keeping every other design variable fixed. This isolates the influence of the attachment topology itself. Lastly, we rerun the prescreening and subsequent simulations to see if these results also apply to the best performing configurations in the prescreening.

6.5.1 Stewart cable routing

The Stewart cable routing is adopted as the reference architecture throughout the remainder of this chapter. The complete numerical results are reported in Tables C.5, C.6, and C.7 in Appendix C.2.2.

Table 6.10 shows the evolution of pose reachability as progressively richer UAV attachment geometries are introduced. The point attachment is unable to reach any target pose, while the aligned, line, disaligned, and diagonal layouts provide only modest improvements. The Stewart configuration reached approximately 65% of the target poses.

Pairwise McNemar tests showed that this improvement in pose reachability is statistically significant for all competing layouts.

A point attachment allows the three UAVs to apply forces through the same location, producing almost no independent rotational authority. Constraining the attachment points to a line or to nearly collinear configurations restores only a subset of the available moments. As the attachment geometry becomes increasingly two-dimensional, the UAV subsystem acquires greater ability to generate arbitrary six-dimensional wrenches. The triangular attachment provides independent moment arms in two dimensions, allowing the UAV subsystem to contribute effectively to the generation of both forces and moments acting on the payload. This increased wrench-generation capability explains the substantial improvement in task completion.

Besides reaching more poses, the Stewart routing also produced the lowest position and orientation errors, and it exhibited the smallest percentage of cable rubbing events (31.7%), indicating that the routing naturally reduces cable interference during the simulated tasks.

The statistical analysis confirms that the Stewart routing provides the highest overall task performance. It achieved the largest mean composite score (5.57), substantially outperforming the diagonal, disaligned and aligned layouts, while the difference with the line configuration was not statistically significant.

From a force-generation perspective, the Stewart configuration achieved the highest capacity margin, the largest worst-case capacity margin, and the greatest radius of the available force set. These results indicate that the architecture maintains a larger safety margin from actuator saturation while preserving greater capability to generate the required payload wrench.

On the other hand, the Stewart routing does not maximize all geometric performance indices. Higher conditioning index and manipulability values were obtained by the aligned and disaligned layouts. Nevertheless, these theoretical advantages did not translate into better execution of the manipulation task, as those configurations exhibited substantially poorer pose tracking and lower success rates. This result highlights that maximizing purely kinematic indices does not necessarily maximize task performance. Although conditioning index and manipulability are useful during preliminary design, they should be interpreted together with task-oriented metrics such as pose reachability, force margins, tracking accuracy, and cable interference.

These results justify selecting the Stewart routing as the baseline architecture for the remainder of the UAV hook-placement study. The following sections therefore investigate the influence of the UAV attachment topology while maintaining the Stewart cable routing fixed.

UAV attachment	**Mean N**	**Mean pose reached**	**Mean cable rubbed**
acts_stewart (triangle)	5.57	65.0%	31.7%
Diagonal	4.00	21.67%	65.0%
Disaligned	4.07	16.67%	68.3%
Line	4.41	6.67%	58.3%
Aligned	2.82	8.33%	66.7%
Point	2.79	0.0%	95.3%

Table 6.10: Mean values of composite score, pose reached and cable rubbed across different drone dispositions for the same ground routing

6.5.2 Ground Screening Across UAV Attachment Topologies

Before evaluating the different UAV attachment topologies, each candidate payload-ground layout was first passed through the same ground-only pre-screening procedure described in Section 6.3.1 and Algorithm 1. Since this stage operates exclusively on the ground-cable Jacobian $J_{p,\text{ground}}$, it is, by construction, independent of the UAV attachment geometry.

This is confirmed by several repeated payload-ground pairs across topologies. `two-triangles-pentagon` is screened under aligned, diagonal, line, and point and returns the identical routing in all four cases, and `two-triangles-rectangle-same` likewise returns the same routing under aligned, diagonal, and line. By contrast, two layouts split into distinct routings depending on the topology under which they are screened. `rhombus-ext-2X3` selects the same routing, under aligned, disaligned, and point, but an alternative routing under diagonal. Similarly, `arrow-rectangle-same` selects different routings under disaligned and line. In every case where the selected routing differs, the indices differ only by a small margin; whenever the routing matches exactly, the indices agree to full precision.

As shown in Table C.3 in Appendix C, the pre-screening returned feasible across all six UAV topologies evaluated (triangle, aligned, diagonal, disaligned, line, and point). As in the Stewart baseline analysis of Section 6.3.1, Wrench Feasibility remains the dominant source of routing rejection, accounting for the large majority of routing-pose failures, while WCC violations are comparatively rare.

The resulting best routings (shown in Table C.4 inside Appendix C) are numerically similar to the triangle baseline reported in Table 6.5: the conditioning index of the selected routing lies in the range $\nu \approx 0.076$-0.083, the smallest singular value in $\sigma_{\min} \approx 0.185$-$0.201$, and manipulability in $w \approx 0.041$-0.096, regardless of which UAV topology the layout is ultimately screened under. Table 6.11 summarizes the best-scoring layout obtained for each topology.

Three of the five non-triangle topologies (aligned, disaligned, and point) converge on the same best layout, `two-triangles-2X3`, with identical indices. This is expected given that all three topologies frequently select identical ground routings for shared payload-ground pairs, as discussed above.

UAV Topology	**Best Layout**	$\sigma_{\min}$	ν	w	$d_{\min}$
Triangle	pentagon - rectangle-same	0.2005	0.0832	0.0721	0.1790
Diagonal	pentagon - rectangle-same	0.2005	0.0832	0.0721	0.1790
Aligned	two-triangles - 2X3	0.1965	0.0820	0.0628	0.0638
Disaligned	two-triangles - 2X3	0.1965	0.0820	0.0628	0.0638
Point	two-triangles - 2X3	0.1965	0.0820	0.0628	0.0638
Line	arrow - rectangle-same	0.1974	0.0819	0.0588	0.3225

Table 6.11: Best-scoring layout obtained during ground screening for each UAV attachment topology.

These pre-screening results alone cannot determine which topology is ultimately preferable for the manipulation task: static wrench-generation indices are informative for eliminating clearly infeasible or singular routings, but they do not reliably predict closed-loop pose reachability, cable rubbing, or tracking accuracy. The values reported here therefore serve to confirm that the ground-only screening stage behaves consistently and as intended across all evaluated UAV topologies, providing a sound basis for the subsequent full-system dynamic evaluation.

6.5.3

Point

The point attachment configuration is never able to command the payload to reach the desired pose, yielding 0 out of 60 successful poses across all setups that employ this drone-placement strategy. This shows that using a shared drone attachment point is a bad design choice, independent from the ground routing it is paired with.

A good way of seeing this limitation is obtained by comparing the same ground cable routing with different drone attachments. For example, contrasting `line-line-triangle` with `line-line-point` shows the reachability dropping from 40.00 % to 0.00%. Cable rubbing is also near-universal (76.67-100%, vs. 31.7% for Stewart) and the capacity margin is negative for every point configuration, meaning the average-pose wrench requirement is not even statically possible with strictly positive tensions.

This is the physically expected consequence of collapsing all three UAV cables onto a single shared attachment point: the drones loose their ability to generate an independent moment about that point, so orientation and fine position control fall almost entirely on the six ground cables. These results show that point-type drone attachment is not an idea and should be left out of the automated architecture search.

Aligned

As shown in Table C.6, `two-triangles-vertical-parallelepiped-aligned` achieves the highest composite score of the whole group $N = 6.9394$, which is about 25% higher than the Stewart baseline (5.5706). However, a p-value of 0.94 shows us how this difference is not statistically significant. Conditioning index and manipulability are significantly better than Stewart for every aligned routing, but this does not translate into closed-loop tracking. When we look at the average of the five ground routings, the aligned attachment only reaches 9.0% of target poses. Despite achieving a mean composite score of 4.61, close to the Stewart value of 5.57, cable rubbing remains extremely frequent, averaging 93.3% across the evaluated routings (Table 6.12).

The aligned case shows how conditioning index and composite score, being static quantities, may overestimate the practical performance of a UAV attachment geometry.

The aligned configuration distributes the UAV attachment points along a single edge of the payload. This geometry improves the conditioning index and manipulability compared with the Stewart layout, indicating a more uniform Jacobian. However, the reduced spatial distribution of the lifting forces limits the ability to generate restoring moments about all axes. As a consequence, despite favorable geometric indices, the dynamic simulations show poor pose reachability and relatively large position and orientation errors.

Disaligned

Like the aligned attachment, the disaligned configuration shows the same qualitative behavior, but in a less extreme way. Conditioning index and manipulability improve significantly over Stewart for all five routings. In contrast, capacity margin and worst-case capacity margin are generally lower, even if not always significantly.

None of the five routings performs well on task completion, succeeding at most 20% of the time. Lastly, cable rubbing dominates.

Line

Although some individual line routings achieve among the highest composite scores of the entire study, these improvements are not representative of the topology as a whole. Averaging across the five evaluated routings, the line attachment achieves a composite score of 5.16, close to the Stewart configuration, but reaches only 4.0% of the target poses while exhibiting an average cable-rubbing rate of 88.3%. For instance, `rectangle-opposite-2x3-line` achieves the highest composite score among all configurations evaluated in this section (8.6194), yet exhibits the one of the lowest pose reachability within the line family (1.67%). This discrepancy further confirms that favorable geometric indices alone are insufficient to characterize the closed-loop performance of the system.

As discussed in Section 6.3.3, constraining attachment points along a single axis inherently limits achievable moment directions, regardless of whether those constrained points lie on the ground or aerial subsystem.

Diagonal

Among the non-triangular UAV attachment topologies, the diagonal configuration provides the best overall dynamic performance. Across its five ground routings, it achieves an average pose reachability of 22.3%, the highest of all the alternative attachment geometries investigated. Nevertheless, this performance remains well below that of the triangular attachment, whose average pose reachability is approximately 65%.

The best individual routing, `rhombus-ext-2x3-diagonal`, reaches 30.0% of the target poses, approximately twice the average best performance obtained with the other non-triangular topologies. Although this is the strongest result within the diagonal family, it still reaches less than half of the poses achieved by the ACTS Stewart baseline. Interestingly, its capacity margin (2.9175) is statistically indistinguishable from that of the Stewart platform (2.7714, p=0.860), while exhibiting a slightly higher nominal value.

As observed for the line and aligned attachment geometries, the static composite score does not reliably predict the dynamic behavior.

6.5.4 Drone disposition comparison analysis

Among all evaluated UAV attachment topologies, the Stewart (triangular) configuration consistently provides the best compromise between static wrench quality and dynamic task execution. It is the only architecture that simultaneously achieves high pose reachability (65.0%), low cable-rubbing rates (31.7%), and accurate position and orientation tracking. Pairwise statistical tests further confirm that its improvement in pose reachability is significant with respect to every alternative attachment topology.

Although several non-triangular topologies obtain comparable values of conditioning index, manipulability, or composite score, these improvements do not translate into successful task execution. In particular, the aligned, disaligned, and line configurations attain average composite scores close to that of the Stewart architecture while reaching only 9.0%, 16.0%, and 4.0% of the target poses, respectively.

These results indicate that the geometric arrangement of the UAV attachment points is substantially more influential than the precise choice of ground routing. Forming a triangle distributes the aerial forces over the payload surface, allowing the UAV subsystem to generate independent forces and moments in all directions while maintaining good cable separation. Consequently, the triangular attachment is adopted as the reference architecture for all subsequent analyses and is retained as the only UAV topology considered during the automated architecture search.

Among the alternative UAV attachment topologies, the diagonal configuration provides the best overall compromise between tracking performance, force-generation capability, and cable interference. It achieves the highest average pose reachability (22.3%) and the lowest cable-rubbing rate (74.0%) among all non-triangular layouts, substantially outperforming the aligned, disaligned, line, and point configurations. Nevertheless, its performance remains well below that of the Stewart architecture.

These observations reinforce the conclusions drawn in Section 6.5.1. While geometric performance indices remain valuable indicators during the preliminary design stage, they cannot alone predict the practical behavior of the system. Metrics directly related to the task, including pose reachability, tracking accuracy and cable interference, provide a more reliable assessment of the overall architecture.

UAV attachment	**Mean composite score**	**Mean pose reached**	**Mean cable rubbed**
Stewart (triangle)	5.57	65.0%	31.7%
Diagonal	4.69	22.3%	74.0%
Disaligned	5.31	16.0%	89.7%
Line	5.16	4.0%	88.3%
Aligned	4.61	9.0%	93.3%
Point	3.86	0.7%	89.0%

Table 6.12: Average composite score, pose reachability, and cable rubbing rate for each UAV attachment topology on the best pre-screening performing

Chapter 7

Conclusions

This thesis investigated the modeling, design, and control of a hybrid Aerial Cable-Towed System (ACTS) intended for collaborative payload manipulation. The work focused on understanding how the architecture of the system, and in particular the placement and routing of the cables, influences its ability to generate the required payload wrench while maintaining stable and accurate motion. Rather than considering the controller independently from the mechanical design, this work addressed the architectural design problem through a systematic methodology combining kinematostatic analysis and dynamic simulation.

A general kinematostatic model of the ACTS was first developed by extending concepts from cable-driven parallel robots to hybrid aerial-ground systems. The model provides a unified framework to evaluate arbitrary configurations of UAV and ground cables and to compute performance metrics related to wrench generation, conditioning, and manipulability. Based on this model, a screening procedure was proposed to efficiently explore the large design space of possible cable routings. Feasibility constraints, including wrench closure, force generation capability, and cable interference, were combined with geometric performance indices to identify a reduced set of promising candidate architectures.

To validate the proposed methodology, a centralized control architecture was implemented in simulation. The controller computes the desired payload wrench, distributes the cable tensions through a quadratic optimization problem, and generates the corresponding UAV commands. A simulation framework based on MuJoCo was developed to evaluate the closed-loop behaviour of the selected architectures under identical operating conditions.

The simulation results demonstrate that the proposed pre-screening successfully removes many unsuitable cable routings while retaining architectures capable of achieving satisfactory dynamic performance. At the same time, they highlight that geometric performance indices alone are insufficient to predict the behaviour of the complete closed-loop system. Architectures exhibiting excellent conditioning and manipulability do not necessarily achieve the highest pose reachability or the lowest tracking errors once payload dynamics, cable tension redistribution, and controller interaction are taken into account. Dynamic simulation therefore represents an essential second stage of the design process, complementing the theoretical analysis rather than replacing it.

Additional studies investigated the influence of cable routing, ground anchor placement, and UAV attachment topology. The results confirm that even when the payload geometry remains unchanged, different cable assignments can produce substantially different dynamic behaviours. The systematic screening methodology consistently produced high-quality candidate routings, while manually designed configurations rarely provided a clear improvement. Furthermore, the analysis of anchor spacing and UAV attachment layouts quantified how these design parameters affect pose reachability, wrench generation capability, and cable interference, providing practical design guidelines for future ACTS implementations.

Overall, this thesis proposes a complete methodology that integrates architectural design, kinematostatic evaluation, controller implementation, and dynamic validation into a unified workflow. The proposed approach reduces the complexity of the architectural design process while providing quantitative criteria for selecting cable configurations suitable for collaborative aerial manipulation. The resulting framework can be readily adapted to different payload geometries, numbers of vehicles, or application scenarios, making it a useful tool for the design of future hybrid aerial cable-driven robotic systems.

Several aspects remain open for future investigation.

First, the methodology should be validated experimentally on a real ACTS platform to assess the influence of sensing uncertainties, communication delays, and unmodeled aerodynamic effects. The current simulations neglect cable elasticity, rotor wash, and environmental disturbances, whose inclusion would improve the realism of the model. The architectural optimization could also be extended to simultaneously optimize cable routing, UAV attachment locations, and ground anchor placement using automated optimization techniques rather than evaluating predefined candidate sets. Finally, integrating trajectory optimization with architecture selection, or adapting the cable routing online through mobile ground anchors or reconfigurable attachment mechanisms, could further improve the adaptability of the system during complex manipulation tasks.

First, this study focuses on architectures composed of six ground cables and three drones. Although this configuration provides full six-degree-of-freedom actuation while limiting the number of aerial vehicles, future work should extend the analysis to different combinations of ground and aerial actuators, investigating how the number of UAVs and ground anchors affects the system performance. A second research direction may include physical interaction with vertical surfaces to assess the influence of contact forces. The proposed methodology also raises several research questions that were outside the scope of this work. One concerns the connection strategy between the UAVs and the payload. In particular, it would be valuable to investigate whether connecting each UAV through one or two cables provides advantages in terms of maneuverability, feasible workspace and available wrench set. Finally, the attachment strategy adopted in this thesis assumes that UAVs are connected to the upper face of the payload while ground anchors are attached to the lower face. Relaxing this assumption could further enlarge the design space. Future investigations should therefore evaluate mixed attachment arrangements, in which aerial and ground cables can be connected to arbitrary payload faces, and assess their influence on workspace and practical usability.

In conclusion, this work demonstrates that the performance of hybrid aerial cable-towed systems depends not only on the control strategy but also on the architectural choices made during the design stage. By combining theoretical analysis with dynamic simulation, the proposed methodology provides a practical and systematic approach for selecting effective ACTS architectures, contributing toward the deployment of cooperative aerial robotic systems capable of performing precise manipulation and interaction tasks in construction and other challenging environments.

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Appendix A

Singularities and Performance Indices

This appendix summarizes the principal singularity types encountered throughout this thesis. In ACTSs, singular or near-singular configurations reduce the system's ability to generate arbitrary payload forces and moments. Different singularities arise from different physical mechanisms.

- Kinematic singularities.

  These occur when the payload Jacobian loses rank ($\text{rank}(J_p) < 6$), reducing the system's ability to independently control all six degrees of freedom. They are typically identified using the smallest singular value of the Jacobian, the conditioning index, or the manipulability measure.

- Wrench singularities.

  These arise when the desired payload wrench lies outside the Available Wrench Set (AWS). Even though the robot may remain kinematically controllable, the required forces and moments cannot be generated while satisfying the cable tension limits. Capacity margin and the radius of the available wrench set are commonly used to quantify the distance from this condition.

- Tension singularities.

  They occur when one or more cable tensions become zero or negative. Since cables can only sustain tensile forces, the affected cables become slack, reducing the system's ability to constrain and control the payload.

Even though these indices are essential to characterize the static capabilities of the robot, they do not completely predict the practical behavior of the system. Dynamic quantities, such as pose reachability, tracking errors, and cable interference, obtained from closed-loop simulations, are also required to comprehensively evaluate an ACTS architecture.

Appendix B
Statistical Methods

This appendix summarizes the statistical methods used to analyze the experimental results presented in Chapter 6. The adopted methodology was selected according to the characteristics of the collected data and the experimental design.

B.1 Hypothesis testing

Statistical hypothesis testing is used to determine whether the experimental observations provide sufficient evidence to support a research claim [25]. Two hypotheses are considered:

- Null hypothesis (H_0): no statistically significant difference exists between the evaluated robot configurations.
- Alternative hypothesis (H_a): at least one robot configuration differs significantly from the others.

The statistical tests produce a p-value, which is the probability of obtaining results at least as extreme as those observed assuming that the null hypothesis is true. In this thesis, a significance level of $\alpha = 0.05$ was adopted, meaning that if $p < 0.05$, the null hypothesis is rejected in favor of the alternative hypothesis. Otherwise, there is not enough evidence to reject the null hypothesis. If the latter happens, it means that there's not enough evidence to conclude that the null hypothesis is false.

B.2 Selection of statistical tests

Every robot configuration was evaluated on the same set of target poses, resulting in paired observations. Non-parametric statistics are used, since several evaluation metrics are binary variables. Therefore, an omnibus test is first performed to determine whether there are significant differences among all evaluated configurations. When this happens, post-hoc pairwise comparisons identify which configurations differ from the selected reference architecture

B.3 Omnibus tests

The Friedman test compares three or more related samples without assuming normally distributed data. In this thesis, it is applied to all continuous performance metrics, including the composite score, conditioning index, manipulability, capacity margin, radius of the available wrench set, and tracking errors.

Cochran's Q test determines whether the probability of success differs among three or more related configurations. It was applied to binary evaluation metrics such as pose reachability, orientation reachability, cable rubbing, and simulation instability.

B.4 Post-hoc pairwise comparisons

Pairwise comparisons are used to identify which configurations differ from the reference architecture. For continuous variables, the Wilcoxon signed-rank test was employed, which compares two related samples by analyzing the signed differences between paired observations and is the non-parametric alternative to the paired Student's t-test. For binary variables, McNemar's exact test was used, which determines whether two related configurations exhibit significantly different probabilities of success.

Throughout this thesis, all post-hoc comparisons were performed against the selected reference configuration.

B.5 Multiple-comparison correction

Performing multiple statistical tests increases the probability of identifying significant differences purely by chance. To reduce this risk, multiple-comparison correction procedures were applied.

While the Benjamini-Hochberg False Discovery Rate (BH-FDR) procedure was used to adjust the omnibus test results, Holm's correction was applied for the post-hoc pairwise comparisons,

Appendix C

Complete Evaluation Results

This appendix collects the complete numerical results supporting the analyses presented in Chapter 6. The appendix is organized according to the two comparative studies performed in the thesis. Section C.1 reports the results of the ground-anchor spacing analysis presented in Section 6.4, whereas Section C.2 contains the pre-screening and simulation results supporting the UAV attachment study of Section 6.5. Within each section, the tables report the complete metric values together with the outcome of the statistical comparisons against the reference Stewart configuration.

The `acts_stewart` platform is used as the reference configuration. Values shown in **bold** indicate statistically significant differences with respect to the reference configuration ($p < 0.05$).

C.1 Ground-anchor spacing study

Configuration	Large	Nominal	Small
Cable Rubbed			
acts_stewart	0.3167	0.3167	0.3167
parallelepiped-int-2x3	0.4333	**0.4667**	**0.6333**
pentagon-rectangle-same-triangle	0.5167	**0.6500**	**0.7833**
rectangle-opposite-2x3	**0.9167**	**0.9333**	**1.0000**
two-triangles-2x3-triangle	**0.8167**	**0.7667**	**0.8000**
two-triangles-rectangle-same-triangle	**1.0000**	**1.0000**	**0.9833**
Capacity Margin γ			
acts_stewart	2.7714	2.7714	2.7714
parallelepiped-int-2x3	**8.1812**	1.7137	**-4.1818**
pentagon-rectangle-same-triangle	**10.5631**	**4.5035**	**1.6203**
rectangle-opposite-2x3	**11.3179**	-0.4414	**-0.6320**
two-triangles-2x3-triangle	**11.2772**	2.2122	**0.9736**
two-triangles-rectangle-same-triangle	**12.0082**	3.4494	0.5723
Composite Score N			
acts_stewart	5.5706	5.5706	5.5706
parallelepiped-int-2x3	**16.7886**	6.7857	**1.7242**
pentagon-rectangle-same-triangle	**18.2912**	6.4366	**3.7686**

Continued on next page

Configuration	Large	Nominal	Small
rectangle-opposite-2x3	**18.9950**	**8.9142**	**4.5529**
two-triangles-2x3-triangle	**17.5658**	6.2580	**3.1418**
two-triangles-rectangle-same-triangle	**21.9579**	**7.3013**	**3.8418**
Conditioning Index ν			
acts_stewart	0.0533	0.0533	0.0533
parallelepiped-int-2x3	**0.1516**	**0.0997**	0.0719
pentagon-rectangle-same-triangle	**0.1547**	**0.0869**	0.0513
rectangle-opposite-2x3	**0.1582**	**0.1246**	**0.0819**
two-triangles-2x3-triangle	**0.1585**	**0.1140**	**0.0682**
two-triangles-rectangle-same-triangle	**0.1596**	**0.1052**	**0.0706**
Manipulability			
acts_stewart	0.1054	0.1054	0.1054
parallelepiped-int-2x3	**1.1789**	**0.4536**	0.2465
pentagon-rectangle-same-triangle	**1.3752**	**0.3522**	0.1269
rectangle-opposite-2x3	**2.1414**	**1.0459**	**0.4775**
two-triangles-2x3-triangle	**1.2135**	**0.5953**	0.2413
two-triangles-rectangle-same-triangle	**1.3245**	**0.4603**	**0.2391**
Radius Available Force			
acts_stewart	11.7301	11.7301	11.7301
parallelepiped-int-2x3	**33.0155**	14.4922	**6.0689**
pentagon-rectangle-same-triangle	**34.2218**	10.9985	**8.2412**
rectangle-opposite-2x3	**36.1480**	**21.5257**	11.8971
two-triangles-2x3-triangle	**29.7475**	11.7302	**6.0347**
two-triangles-rectangle-same-triangle	**40.5137**	13.8305	**7.9175**
Worst Case Capacity Margin			
acts_stewart	0.7337	0.7337	0.7337
parallelepiped-int-2x3	0.6786	0.6981	0.3715
pentagon-rectangle-same-triangle	**0.7415**	**0.7388**	0.5331
rectangle-opposite-2x3	**0.7385**	-0.4904	-0.1371
two-triangles-2x3-triangle	**0.7403**	0.6680	0.6332
two-triangles-rectangle-same-triangle	**0.7715**	**0.5969**	0.6017

Table C.1: Static performance metrics

Configuration	Large	Nominal	Small
Orientation Error			
acts_stewart	0.1223	0.1223	0.1223
parallelepiped-int-2x3	0.1945	0.3063	**0.5346**
pentagon-rectangle-same-triangle	0.1277	0.0457	0.2215
rectangle-opposite-2x3	0.2831	0.3855	**0.5682**
two-triangles-2x3-triangle	0.1961	0.3396	0.3975
two-triangles-rectangle-same-triangle	0.1079	0.3338	0.3862
Position Error			
acts_stewart	0.0592	0.0592	0.0592
parallelepiped-int-2x3	0.5278	0.8452	**1.6011**
pentagon-rectangle-same-triangle	0.1030	0.0555	**0.2144**
rectangle-opposite-2x3	**0.3350**	**1.4925**	**1.7521**
two-triangles-2x3-triangle	0.0910	**1.0747**	**0.8745**
two-triangles-rectangle-same-triangle	0.0505	0.5184	**0.9800**
Pose Reached			
acts_stewart	0.6500	0.6500	0.6500
parallelepiped-int-2x3	0.7667	0.7500	0.5667
pentagon-rectangle-same-triangle	0.8333	0.8333	0.6000
rectangle-opposite-2x3	0.6500	0.5833	0.4333
two-triangles-2x3-triangle	0.7667	0.6667	0.6667
two-triangles-rectangle-same-triangle	0.7500	0.6500	0.6000
Simulation Unstable			
acts_stewart	0.0833	0.0833	0.0833
parallelepiped-int-2x3	0.0500	0.0500	0.0833
pentagon-rectangle-same-triangle	0.0167	0.0000	0.0333
rectangle-opposite-2x3	0.0667	0.0500	0.1500
two-triangles-2x3-triangle	0.0667	0.0833	0.0500
two-triangles-rectangle-same-triangle	0.1000	0.1833	0.0667

Table C.2: Dynamic performance metrics

C.2 UAV attachment study

C.2.1 Pre-screening results

Configuration	WCC Failures	WFC Failures	IFC Failures	Feasible
Aligned Configurations				
two-triangles-2X3-aligned	72	2315	485	✓
two-triangles-pentagon-aligned	0	1296	249	✓
two-triangles-vertical-parallelepiped-aligned	64	2555	507	✓
rhombus-ext-2X3-aligned	52	296	97	✓
two-triangles-rectangle-same-aligned	6	792	110	✓
Diagonal Configurations				
pentagon-rectangle-same-diagonal	0	400	42	✓
two-triangles-pentagon-diagonal	0	1290	249	✓
two-triangles-rectangle-same-diagonal	6	790	110	✓
rhombus-int-rectangle-same-diagonal	8	74	19	✓
rhombus-ext-2X3-diagonal	52	301	97	✓
Disaligned Configurations				
two-triangles-2X3-disaligned	72	2313	485	✓
arrow-rectangle-same-disaligned	12	378	46	✓
two-triangles-vertical-parallelepiped-disaligned	64	2553	507	✓
rhombus-ext-2X3-disaligned	52	301	97	✓
rectangle-opposite-2X3-disaligned	72	146	94	✓
Line Configurations				
arrow-rectangle-same-line	12	379	46	✓
two-triangles-pentagon-line	0	1295	249	✓
two-triangles-rectangle-same-line	6	790	110	✓
2X3-rectangle-same-line	60	146	106	✓
rectangle-opposite-2X3-line	72	148	94	✓
Point Configurations				
two-triangles-2X3-point	72	2328	485	✓
two-triangles-pentagon-point	0	1301	249	✓
two-triangles-vertical-parallelepiped-point	64	2567	507	✓
rhombus-ext-2X3-point	52	305	97	✓
rhombus-int-rectangle-same-point	8	75	19	✓

Table C.3: Pre-screening results

Configuration	$\sigma_{\min}$ [a]	ν [b]	w [c]	$d_{\min}$ [d]
Aligned Configurations				
two-triangles-2X3-aligned	0.1965	0.0820	0.0628	0.0638
two-triangles-pentagon-aligned	0.1979	0.0818	0.0429	0.0836
two-triangles-vertical-parallelepiped-aligned	0.1911	0.0797	0.0450	0.0638
rhombus-ext-2X3-aligned	0.1925	0.0793	0.0406	0.2057
two-triangles-rectangle-same-aligned	0.1893	0.0793	0.0740	0.0982
Diagonal Configurations				
pentagon-rectangle-same-diagonal	0.2005	0.0832	0.0721	0.1790
two-triangles-pentagon-diagonal	0.1979	0.0818	0.0429	0.0836
two-triangles-rectangle-same-diagonal	0.1893	0.0793	0.0740	0.0982
rhombus-int-rectangle-same-diagonal	0.1863	0.0773	0.0521	0.2121
rhombus-ext-2X3-diagonal	0.1854	0.0764	0.0405	0.2399
Disaligned Configurations				
two-triangles-2X3-disaligned	0.1965	0.0820	0.0628	0.0638
arrow-rectangle-same-disaligned	0.1959	0.0812	0.0587	0.1913
two-triangles-vertical-parallelepiped-disaligned	0.1911	0.0797	0.0450	0.0638
rhombus-ext-2X3-disaligned	0.1925	0.0793	0.0406	0.2057
rectangle-opposite-2X3-disaligned	0.1875	0.0786	0.0957	0.2027
Line Configurations				
arrow-rectangle-same-line	0.1974	0.0819	0.0588	0.3225
two-triangles-pentagon-line	0.1979	0.0818	0.0429	0.0836
two-triangles-rectangle-same-line	0.1893	0.0793	0.0740	0.0982
2X3-rectangle-same-line	0.1885	0.0790	0.0896	0.2721
rectangle-opposite-2X3-line	0.1875	0.0786	0.0957	0.2027
Point Configurations				
two-triangles-2X3-point	0.1965	0.0820	0.0628	0.0638
two-triangles-pentagon-point	0.1979	0.0818	0.0429	0.0836
two-triangles-vertical-parallelepiped-point	0.1911	0.0797	0.0450	0.0638
rhombus-ext-2X3-point	0.1925	0.0793	0.0406	0.2057
rhombus-int-rectangle-same-point	0.1863	0.0773	0.0521	0.2121

Table C.4: Selected routing metrics

[a] $\sigma_{\min}$ = smallest singular value of the payload Jacobian.
[b] ν = conditioning index (Eq. 3.29)
[c] w = manipulability (Eq. 3.30)
[d] $d_{\min}$ = minimum cable-to-cable clearance of the selected routing over all tested poses.

C.2.2 Simulation results

Configuration	Position Error	Orien-tation Error	Pose Reached Rate	Simulation Unstable Rate
acts_stewart	0.0592	0.1223	65.00%	8.33%
Aligned Configurations				
rhombus-ext-2x3-aligned	**0.9787**	**0.5777**	**28.33%**	11.67%
two-triangles-pentagon-aligned	**2.2666**	**0.9380**	**8.33%**	5.00%
two-triangles-2x3-aligned	**2.6047**	**1.5192**	**1.67%**	6.67%
two-triangles-vertical-parallelepiped-aligned	**3.0221**	**1.3568**	**1.67%**	16.67%
two-triangles-rectangle-same-aligned	**3.0815**	**1.3531**	**5.00%**	10.00%
Diagonal Configurations				
rhombus-ext-2x3-diagonal	**0.3978**	**0.2576**	**30.00%**	0.00%
rhombus-int-rectangle-same-diagonal	**0.6038**	**0.3671**	**28.33%**	5.00%
pentagon-rectangle-same-diagonal	**1.2065**	**0.7766**	**15.00%**	8.33%
two-triangles-pentagon-diagonal	**1.2651**	**0.5569**	**18.33%**	3.33%
two-triangles-rectangle-same-diagonal	**1.3362**	**0.5705**	**20.00%**	15.00%
Disaligned Configurations				
rhombus-ext-2x3-disaligned	**1.1077**	**0.7734**	**23.33%**	3.33%
two-triangles-vertical-parallelepiped-disaligned	**1.1213**	**0.7439**	**16.67%**	15.00%
two-triangles-2x3-disaligned	**0.6524**	**0.5183**	**16.67%**	13.33%
rectangle-opposite-2x3-disaligned	**2.6658**	**1.1580**	**3.33%**	11.67%
arrow-rectangle-same-disaligned	**0.6461**	**0.5435**	**20.00%**	6.67%
Line Configurations				
rectangle-opposite-2x3-line	**3.0277**	**1.3840**	**1.67%**	8.33%
arrow-rectangle-same-line	**0.5130**	**0.5029**	**10.00%**	3.33%
2x3-rectangle-same-line	**2.0849**	**1.2964**	**0.00%**	8.33%
two-triangles-pentagon-line	**2.0287**	**0.8962**	**5.00%**	3.33%
two-triangles-rectangle-same-line	**1.9877**	**1.3895**	**3.33%**	13.33%
Point Configurations				
rhombus-ext-2x3-point	**2.6894**	**0.6838**	**0.00%**	8.33%
two-triangles-2x3-point	**2.8918**	**1.0663**	**0.00%**	11.67%
rhombus-int-rectangle-same-point	**2.4764**	**0.7987**	**0.00%**	0.00%
two-triangles-pentagon-point	**2.8700**	**0.9154**	**1.67%**	5.00%
two-triangles-vertical-parallelepiped-point	**2.8318**	**1.4063**	**1.67%**	10.00%

Table C.5: Tracking performance.

Configuration	Composite Score	Cable Rubbed Rate	Capacity Margin	Worst Case Margin
acts_stewart	5.5706	31.67%	2.7714	0.7337
Aligned Configurations				
two-triangles-rectangle-same-aligned	4.7392	**100.00%**	**-6.0528**	**0.2226**
rhombus-ext-2x3-aligned	4.4362	**75.00%**	-0.9302	**0.5738**
two-triangles-pentagon-aligned	**3.7571**	**93.33%**	**-3.6281**	**0.1000**
two-triangles-vertical-parallelepiped-aligned	6.9394	**100.00%**	**-3.6846**	**0.0807**
two-triangles-2x3-aligned	**3.1612**	**98.33%**	**-3.4437**	**0.1422**
Diagonal Configurations				
rhombus-ext-2x3-diagonal	4.2652	**50.00%**	2.9175	0.6638
two-triangles-rectangle-same-diagonal	5.8950	**100.00%**	0.8568	0.6000
pentagon-rectangle-same-diagonal	5.0529	**76.67%**	1.6229	**0.5329**
rhombus-int-rectangle-same-diagonal	4.5522	**61.67%**	1.4127	0.6701
two-triangles-pentagon-diagonal	**3.6751**	**81.67%**	1.2228	**0.5915**
Disaligned Configurations				
rhombus-ext-2x3-disaligned	**3.8389**	**81.67%**	0.5601	**0.5143**
two-triangles-vertical-parallelepiped-disaligned	5.2415	**96.67%**	-0.2683	**0.5776**
two-triangles-2x3-disaligned	7.0001	**95.00%**	3.3010	**0.6088**
rectangle-opposite-2x3-disaligned	6.5342	**96.67%**	**-3.8850**	**-0.5714**
arrow-rectangle-same-disaligned	3.9368	**78.33%**	2.4076	**0.6155**
Line Configurations				
rectangle-opposite-2x3-line	8.6194	**100.00%**	**-3.4354**	**-0.0719**
arrow-rectangle-same-line	5.2339	**60.00%**	2.8460	**0.5861**
2x3-rectangle-same-line	4.2267	**98.33%**	**-5.4921**	**-0.2952**
two-triangles-pentagon-line	**3.9568**	**83.33%**	**-1.6696**	**0.4857**
two-triangles-rectangle-same-line	**3.7605**	**100.00%**	**-4.2023**	**0.2509**
Point Configurations				
rhombus-ext-2x3-point	5.0544	**76.67%**	**-0.0824**	**0.2626**
two-triangles-2x3-point	**2.9828**	**100.00%**	**-9.4867**	**-0.9989**
rhombus-int-rectangle-same-point	5.0534	**78.33%**	**-4.0811**	**0.3515**
two-triangles-pentagon-point	4.4593	**90.00%**	**-2.6007**	**0.2692**
two-triangles-vertical-parallelepiped-point	**1.7512**	**100.00%**	**-9.6731**	**-1.9053**

Table C.6: Task performance.

Configuration	**Conditioning Index**	**Manipulability**	**Radius Available Force**
acts_stewart	0.0533	0.1054	11.7301
Aligned Configurations			
two-triangles-pentagon-aligned	**0.1345**	**0.9516**	**10.1458**
two-triangles-vertical-parallelepiped-aligned	**0.1274**	**0.8604**	18.5554
two-triangles-2x3-aligned	**0.1187**	**0.8218**	**9.6870**
two-triangles-rectangle-same-aligned	**0.1179**	**0.9361**	14.7220
rhombus-ext-2x3-aligned	**0.0915**	**0.4400**	**9.5317**
Diagonal Configurations			
pentagon-rectangle-same-diagonal	**0.1041**	**0.7107**	9.7685
two-triangles-pentagon-diagonal	**0.1031**	**0.5257**	**7.1658**
two-triangles-rectangle-same-diagonal	**0.1017**	**0.5136**	12.4682
rhombus-int-rectangle-same-diagonal	**0.0874**	**0.3473**	**9.0086**
rhombus-ext-2x3-diagonal	**0.0822**	**0.2362**	**6.7424**
Disaligned Configurations			
rhombus-ext-2x3-disaligned	**0.0872**	**0.3856**	**7.5565**
two-triangles-vertical-parallelepiped-disaligned	**0.0957**	**0.3979**	11.4287
two-triangles-2x3-disaligned	**0.1032**	**0.4329**	12.8803
rectangle-opposite-2x3-disaligned	**0.1169**	**1.3685**	19.7337
arrow-rectangle-same-disaligned	**0.0931**	**0.4585**	**6.8325**
Line Configurations			
rectangle-opposite-2x3-line	**0.1280**	**1.2408**	25.5407
arrow-rectangle-same-line	**0.0889**	**0.3538**	9.8717
2x3-rectangle-same-line	**0.1223**	**0.8881**	14.0758
two-triangles-pentagon-line	**0.1130**	**0.6483**	10.1968
two-triangles-rectangle-same-line	**0.1050**	**0.6833**	11.3189
Point Configurations			
rhombus-ext-2x3-point	**0.1488**	**1.2594**	9.9832
two-triangles-2x3-point	**0.1248**	**0.8800**	12.4516
rhombus-int-rectangle-same-point	**0.1279**	**0.8235**	13.6485
two-triangles-pentagon-point	**0.1454**	**1.0357**	11.0253
two-triangles-vertical-parallelepiped-point	**0.1054**	**0.5491**	9.7853

Table C.7: Geometric performance metrics.